

penev_A — formula report

674 inline formulas, 116 display equations, 1 tables, 25 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
penev_A_ E00041 (2.44)	008	■ 0.244	$\begin{aligned} \Phi &= \sum_r \sigma_r \mathbf{s}_r (\psi_r,) \\ \Phi^* \Phi &= \mathbf{R} = \sum_r \sigma_r^2 (\psi_r,) \\ \mathrm{tr} \mathbf{R} &= \sum_r \sigma_r^2. \end{aligned}$	$\begin{aligned} \Phi &= \sum_r \sigma_r \mathbf{s}_r (\psi_r,) \\ \Phi^* \Phi &= \mathbf{R} = \sum_r \sigma_r^2 (\psi_r,) \\ \mathrm{tr} \mathbf{R} &= \sum_r \sigma_r^2. \end{aligned}$	
penev_A_E00004 (2.4)	006	■ 0.303	$\begin{aligned} &\left(\hat{\mathbf{x}}, \hat{\mathbf{x}}'\right)_V \equiv V \delta_{\mathbf{x}, \mathbf{x}'} \\ &\left\langle \hat{t}, \hat{t}' \right\rangle_S \equiv T \delta_{t, t'} \end{aligned}$	$\begin{aligned} (\hat{\mathbf{x}}, \hat{\mathbf{x}}')_V &\equiv V \delta_{\mathbf{x}, \mathbf{x}'} \\ \langle \hat{t}, \hat{t}' \rangle_S &\equiv T \delta_{t, t'} \end{aligned}$	
penev_A_ E00075 (4.17)	018	■ 0.314	$\begin{aligned} \mathrm{0} &= \sum_f \mathrm{s_f} (\Psi_f,) \\ \mathrm{0}^* &= \sum_f \Psi_f \left\langle \mathrm{s_f}, \right\rangle. \end{aligned}$	$\begin{aligned} \mathbf{O} &= \sum_f \mathbf{s_f} (\Psi_f,) \\ \mathbf{O}^* &= \sum_f \Psi_f \langle \mathbf{s_f}, \rangle. \end{aligned}$	
penev_A_ E00030 (2.33)	007	■ 0.397	$\Phi^{*+} \Phi^* = \mathbf{P}_{\mathrm{range}(\Phi)} = \mathbf{P}' = \sum_{\sigma>0} \mathbf{s}_{\sigma} \left\langle \mathbf{s}_{\sigma}, \right\rangle.$	$\Phi^{*+} \Phi^* = \mathbf{P}_{\mathrm{range}(\Phi)} = \mathbf{P}' = \sum_{\sigma>0} \mathbf{s}_{\sigma} \langle \mathbf{s}_{\sigma}, \rangle.$	
penev_A_ E00039 (2.42)	008	■ 0.403	$\left\langle \Phi \psi_{\sigma}, \Phi \psi_{\sigma'} \right\rangle = \sigma \sigma' \frac{1}{T} \sum_t a_{\sigma}^t a_{\sigma'}^t = \sigma^2 \delta_{\sigma_1 \sigma'}$	$\langle \Phi \psi_{\sigma}, \Phi \psi_{\sigma'} \rangle = \sigma \sigma' \frac{1}{T} \sum_t a_{\sigma}^t a_{\sigma'}^t = \sigma^2 \delta_{\sigma, \sigma'}.$	
penev_A_ E00035 (2.48)	008	■ 0.409	$\left\ \mathbf{P}' \hat{t} \right\ _S^2 = \left\langle \mathbf{P}' \hat{t}, \mathbf{P}' \hat{t} \right\rangle = \sum_{\sigma>0} \left\langle \mathbf{s}_{\sigma}, \hat{t} \right\rangle^2 = \sum_{\sigma>0} u_{\sigma}^{t^2}.$	$\ \mathbf{P}' \hat{t} \ _S^2 = \langle \mathbf{P}' \hat{t}, \mathbf{P}' \hat{t} \rangle = \sum_{\sigma>0} \langle \mathbf{s}_{\sigma}, \hat{t} \rangle^2 = \sum_{\sigma>0} u_{\sigma}^{t^2}.$	
penev_A_E00007	019	■ 0.429	$\begin{aligned} K^{\{1\}} &= \mathrm{K} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^1 (\psi_r,) \\ K^{(0)} &= P = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^0 (\psi_r,) \\ K^{(-1)} &= K^{(-1)} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-1} (\psi_r,) \end{aligned}$	$\begin{aligned} \mathbf{K}^{(1)} &= \mathbf{K} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^1 (\psi_r,) \\ \mathbf{K}^{(0)} &= \mathbf{P} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^0 (\psi_r,) \\ \mathbf{K}^{(-1)} &= \mathbf{K}^{(-1)} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-1} (\psi_r,) \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					

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penev_A_EQ0055 (3.6)	014	■ 0.496	\begin{aligned} F_{\text{r}}^{\text{same}}(u) &= \mathcal{P}\left[D_{\text{r}}^{(p,q)} \leq u\right]_{(p,q) \in C_{\text{aame}}} \\ F_{\text{r}}^{\text{diff}}(u) &= \mathcal{P}\left[D_{\text{r}}^{(p,q)} \leq u\right]_{(p,q) \in C_{\text{diff}}} \end{aligned}	$F_r^{\text{same}}(u) = \mathcal{P}\left[D_r^{(p,q)} \leq u\right]_{(p,q) \in C_{\text{aame}}}$ $F_r^{\text{diff}}(u) = \mathcal{P}\left[D_r^{(p,q)} \leq u\right]_{(p,q) \in C_{\text{diff}}}$	$F_r^{\text{same}}(u) = \mathcal{P}[D_r^{(p,q)} \leq u]_{(p,q) \in C_{\text{aame}}}$ $F_r^{\text{diff}}(u) = \mathcal{P}[D_r^{(p,q)} \leq u]_{(p,q) \in C_{\text{diff}}}$
penev_A_EQ0070 (4.12)	018	■ 0.502	$\mathrm{0}^+ \mathbf{0} = \mathbf{0}$	$\mathbf{O}^+ \mathbf{O} = \mathbf{1}_U$.	$\mathbf{O}^* \mathbf{O} = \mathbf{1}_U$.
penev_A_EQ0084 (4.26)	019	■ 0.525	\begin{aligned} \mathrm{K} &= \mathrm{0}^* \mathrm{\Phi}^{*+} = \sum_{\mathrm{f}} \Psi_{\mathrm{f}} \left(\frac{1}{\sigma_{\mathrm{f}}} \right)^1 (\Psi_{\mathrm{f}},) \\ 1 &= \mathrm{O}^* \mathrm{O} = \sum_{\mathrm{f}} \Psi_{\mathrm{f}} \left(\frac{1}{\sigma_{\mathrm{f}}} \right)^0 (\Psi_{\mathrm{f}},) \\ \mathrm{K}^{-1} &= \mathrm{\Phi}^* \mathrm{O}^{*+} = \sum_{\mathrm{f}} \Psi_{\mathrm{f}} \left(\frac{1}{\sigma_{\mathrm{f}}} \right)^{-1} (\Psi_{\mathrm{f}},) \\ \mathrm{R} &= \mathrm{\Phi}^* \mathrm{\Phi} = \sum_{\mathrm{f}} \Psi_{\mathrm{f}} \left(\frac{1}{\sigma_{\mathrm{f}}} \right)^{-2} (\Psi_{\mathrm{f}},) \end{aligned}	$\mathbf{K} = \mathbf{O}^* \Phi^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^1 (\Psi_{\mathbf{f}},)$ $1 = \mathbf{O}^* \mathbf{O} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^0 (\Psi_{\mathbf{f}},)$ $\mathbf{K}^{-1} = \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-1} (\Psi_{\mathbf{f}},)$ $\mathbf{R} = \Phi^* \Phi = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-2} (\Psi_{\mathbf{f}},)$	$\mathbf{K} = \mathbf{O}^* \Phi^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^1 (\Psi_{\mathbf{f}},)$ $1 = \mathbf{O}^* \mathbf{O} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^0 (\Psi_{\mathbf{f}},)$ $\mathbf{K}^{-1} = \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-1} (\Psi_{\mathbf{f}},)$ $\mathbf{R} = \Phi^* \Phi = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{-2} (\Psi_{\mathbf{f}},)$
penev_A_EQ0021 (2.24)	007	■ 0.544	$\mathrm{\Phi}^* \mathrm{\Phi} \psi_{\sigma} \equiv \mathbf{R} \psi_{\sigma} = \sigma^2 \psi_{\sigma}$ with $(\psi_{\sigma}, \psi_{\sigma'}) = \delta_{\sigma, \sigma'}$	$\Phi^* \Phi \psi_{\sigma} \equiv \mathbf{R} \psi_{\sigma} = \sigma^2 \psi_{\sigma}$ with $(\psi_{\sigma}, \psi_{\sigma'}) = \delta_{\sigma, \sigma'}$	$\Phi^* \Phi \psi_{\sigma} \equiv \mathbf{R} \psi_{\sigma} = \sigma^2 \psi_{\sigma}$ with $(\psi_{\sigma}, \psi_{\sigma'}) = \delta_{\sigma, \sigma'}$
penev_A_EQ0094 (4.35)	020	■ 0.552	$\phi = \mathbf{P}_N \phi = \Phi^* \Phi^{*+} \phi = \Phi^* \mathbf{O}^{*+} O = \mathbf{K}^{(-1)} O = \sum_{r=1}^N a_r \sigma_r \psi_r$	$\phi = \mathbf{P}_N \phi = \Phi^* \Phi^{*+} \phi = \Phi^* \mathbf{O}^{*+} O = \mathbf{K}^{(-1)} O = \sum_{r=1}^N a_r \sigma_r \psi_r$	$\phi = \mathbf{P}_N \phi = \Phi^* \Phi^{*+} \phi = \Phi^* \mathbf{O}^{*+} O = \mathbf{K}^{(-1)} O = \sum_{r=1}^N a_r \sigma_r \psi_r.$
penev_A_EQ0057 (3.8)	015	■ 0.569	$w_t^{Fisher} \propto \sum_{r=1}^N (m_r^{sqame} - m_r^{diff}) (H^{intmg})_{rt}^{-1}$	$w_t^{Fisher} \propto \sum_{r=1}^N (m_r^{sqame} - m_r^{diff}) (H^{intmg})_{rt}^{-1}$	$w_t^{Fisher} \propto \sum_{r=1}^N (m_r^{sqame} - m_r^{diff}) (H^{intmg})_{rt}^{-1}$
penev_A_EQ0069 (4.11)	018	■ 0.571	$(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = V \delta_{\mathbf{x}, \mathbf{y}} = \frac{1}{T} \sum_t O^t(\mathbf{x}) O^t(\mathbf{y}) = \langle \mathbf{O} \hat{\mathbf{x}}, \mathbf{O} \hat{\mathbf{y}} \rangle = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{y}})$	$(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = V \delta_{\mathbf{x}, \mathbf{y}} = \frac{1}{T} \sum_t O^t(\mathbf{x}) O^t(\mathbf{y}) = \langle \mathbf{O} \hat{\mathbf{x}}, \mathbf{O} \hat{\mathbf{y}} \rangle = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{y}})$	$(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = V \delta_{\mathbf{x}, \mathbf{y}} = \frac{1}{T} \sum_{\mathbf{t}} O^t(\mathbf{x}) O^t(\mathbf{y}) = \langle \mathbf{O} \hat{\mathbf{x}}, \mathbf{O} \hat{\mathbf{y}} \rangle = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{y}})$
penev_A_EQ0111 (4.46)	022	■ 0.585	$\mathbf{q}_m \equiv \mathbf{q}_{\mathbf{x}_m} = \mathbf{O} \hat{\mathbf{x}}_m$.	$\mathbf{q}_m \equiv \mathbf{q}_{\mathbf{x}_m} = \mathbf{O} \hat{\mathbf{x}}_m$.	$\mathbf{q}_m \equiv \mathbf{q}_{\mathbf{x}_m} = \mathbf{O} \hat{\mathbf{x}}_m$.
penev_A_EQ0061 (4.3)	017	■ 0.590	$\Psi_{\mathbf{f}}(\mathbf{x}) = e^{\mathbf{if} \cdot \mathbf{x}}$.	$\Psi_{\mathbf{f}}(\mathbf{x}) = e^{\mathbf{if} \cdot \mathbf{x}}$.	$\Psi_{\mathbf{f}}(\mathbf{x}) = e^{\mathbf{if} \cdot \mathbf{x}}$.
penev_A_EQ0106 (4.47)	022	■ 0.622	$(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}) = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{x}}_m) = (\hat{\mathbf{x}}, \mathbf{P} \hat{\mathbf{x}}_m) \equiv (\hat{\mathbf{x}}, \mathbf{P}_{\mathbf{x}_m}) \equiv P_{\mathbf{x}_m}(\mathbf{x})$.	$(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}) = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{x}}_m) = (\hat{\mathbf{x}}, \mathbf{P} \hat{\mathbf{x}}_m) \equiv (\hat{\mathbf{x}}, \mathbf{P}_{\mathbf{x}_m}) \equiv P_{\mathbf{x}_m}(\mathbf{x})$.	$(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}) = (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{O} \hat{\mathbf{x}}_m) = (\hat{\mathbf{x}}, \mathbf{P} \hat{\mathbf{x}}_m) \equiv (\hat{\mathbf{x}}, \mathbf{P}_{\mathbf{x}_m}) \equiv P_{\mathbf{x}_m}(\mathbf{x})$.
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					

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penev_A_ EQ0018 (2.21)	007	0.644	$\left(\mathrm{u}^{\prime},\Phi^{*}\Phi\mathrm{u}\right)=\left\langle\Phi\mathrm{u}^{\prime},\Phi\mathrm{u}\right\rangle=\left(\Phi^{*}\Phi\mathrm{u}^{\prime},\mathrm{u}\right).$ $\left(\mathrm{u}^{\prime},\Phi^{*}\Phi\mathrm{u}\right)=\left\langle\Phi\mathrm{u}^{\prime},\Phi\mathrm{u}\right\rangle=\left(\Phi^{*}\Phi\mathrm{u}^{\prime},\mathrm{u}\right).$		
penev_A_ EQ0077 (4.19)	018	0.654	$\mathrm{K}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\frac{1}{\sigma_{\mathbf{f}}}(\Psi_{\mathbf{f}},)=0^{*}\Phi^{*+}$		$\mathbf{K}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\frac{1}{\sigma_{\mathbf{f}}}(\Psi_{\mathbf{f}},)=\mathbf{0}^{*}\Phi^{*+}.$
penev_A_ EQ0092 (4.33)	020	0.657	$0^{*+}O=\Phi^{*+}\phi$		$\mathbf{0}^{*+}O=\Phi^{*+}\phi$
penev_A_ EQ0096 (4.37)	021	0.658	$O_{\{m\}}=\left\langle\mathrm{q}_{\mathbf{x}_m},\mathbf{s}\right\rangle$		$O_m=\langle\mathbf{q}_{\mathbf{x}_m},\mathfrak{s}\rangle$
penev_A_EQ0063 (4.5)	017	0.666	$\Phi^{*}\Phi=\mathbf{R}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}\sigma_{\mathbf{f}}^2(\Psi_{\mathbf{f}},)$		$\Phi^{*}\Phi=\mathbf{R}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}o_{\mathbf{f}}^2(\Psi_{\mathbf{f}},)$
penev_A_ EQ0112 (4.53)	022	0.692	$\mathbf{p}_l=\sum_{n=1}^{ M }\mathbf{B}_{ln}\mathbf{q}_n.$		$\mathbf{p}_{\mathfrak{l}}=\sum_{n=1}^{ \mathcal{M} }\mathbf{B}_{ln}\mathbf{q}_n.$
penev_A_ EQ0026 (2.29)	007	0.693	$\left\langle\mathbf{s}_{\sigma},\mathbf{s}_{\sigma'}\right\rangle=\delta_{\sigma,\sigma'}$		$\langle\mathfrak{s}_{\sigma},\mathfrak{s}_{\sigma'}\rangle=\delta_{\sigma,\sigma'}$
penev_A_ EQ0113 (4.54)	022	0.695	$\left\langle\mathbf{q}_m,\mathbf{p}_i\right\rangle=\delta_{m,i}.$		$\langle\mathbf{q}_m,\mathbf{p}_i\rangle=\delta_{m,i}.$
penev_A_ EQ0071 (4.13)	018	0.710	$O^{*}O=\sum_f\Psi_f(\Psi_f,)$		$\mathbf{0}^{*}\mathbf{O}=\sum_{\mathbf{f}}\Psi_{\mathbf{f}}(\Psi_{\mathbf{f}},).$
penev_A_ EQ0031 (2.34)	007	0.715	$\Phi^{*}\Phi^{*+}=\mathbf{P}_{\mathrm{range}}\left(\Phi^{*}\right)=\mathbf{P}=\sum_{\sigma>0}\psi_{\sigma}(\psi_{\sigma},).$		$\Phi^{*}\Phi^{*+}=\mathbf{P}_{\mathrm{range}(\Phi^{*})}=\mathbf{P}=\sum_{\sigma>0}\psi_{\sigma}(\psi_{\sigma},).$
penev_A_ EQ0028 (2.31)	007	0.721	$\Phi^{*}=\sum_{\sigma>0}\sigma\psi_{\sigma}\left\langle\mathbf{s}_{\sigma},\right\rangle.$		$\Phi^{*}=\sum_{\sigma>0}\sigma\psi_{\sigma}(\mathfrak{s}_{\sigma},).$
penev_A_ EQ0107 (4.48)	022	0.724	$O^{rec}(\mathbf{x})=O_ma_{\mathbf{x}_{\mathbf{wa}}}(\mathbf{x})$		$O^{rec}(\mathbf{x})=O_ma_{\mathbf{x}_m}(\mathbf{x})$
penev_A_ EQ0040 (2.43)	008	0.742	$\mathcal{P}\left[\phi^t\right]=\mathcal{P}\left[\mathbf{P}'\hat{t}\right]=\prod_{\sigma>0}\mathcal{P}\left[a_{\sigma}^t\right].$		$\mathcal{P}[\phi^t]=\mathcal{P}[\mathbf{P}'\hat{t}]=\prod_{\sigma>0}\mathcal{P}[a_{\sigma}^t].$
penev_A_ EQ0100 (4.41)	022	0.748	$\left\langle q_{x_m},p_{x_m}\right\rangle=1$		$\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{p}_{\mathbf{x}_m}\rangle=1$
penev_A_ EQ0073 (4.15)	018	0.789	$O^{*}o_{\mathbf{f}}=\Psi_f.$		$\mathbf{0}^{*}\mathbf{o}_{\mathbf{f}}=\Psi_{\mathbf{f}}.$
penev_A_ EQ0104 (4.45)	022	0.789	$O^{rec}(\mathbf{x})=\left(\hat{\mathbf{x}},\mathbf{0}^{*}\mathbf{s}^{rec}\right).$		$O^{rec}(\mathbf{x})=\langle\hat{\mathbf{x}},\mathbf{0}^{*}\mathfrak{s}^{rec}\rangle.$
penev_A_ EQ0047 (2.50)	008	0.799	$\phi_N^{\mathrm{rec}}=(\Phi_N)^*(\Phi_N)^{*+}\phi=\mathbf{P}_N\phi=\sum_{r=1}^Na_r\sigma_r\psi_r$		$\phi_N^{\mathrm{rec}}=(\Phi_N)^*(\Phi_N)^{*+}\phi=\mathbf{P}_N\phi=\sum_{r=1}^Na_r\sigma_r\psi_r$
penev_A_ EQ0034 (2.37)	008	0.801	$\mathbf{P}'\hat{t}=\sum_{\sigma>0}\mathbf{s}_{\sigma}\left\langle\mathbf{s}_{\sigma},\hat{t}\right\rangle\equiv\sum_{\sigma>0}a_{\sigma}^i\mathbf{s}_{\sigma}$		$\mathbf{P}'\hat{t}=\sum_{\sigma>0}\mathfrak{s}_{\sigma}(\mathfrak{s}_{\sigma},\hat{t})\equiv\sum_{\sigma>0}a_{\sigma}^i\mathfrak{s}_{\sigma}$
penev_A_ EQ0074 (4.16)	018	0.811	$o_f=s_f.$		$\mathbf{o}_{\mathbf{f}}=\mathfrak{s}_{\mathbf{f}}.$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

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penev_A_ EQ0079 (4.21)	018	■0.822	$\begin{aligned} \left\langle \Phi^{*+} \right \Phi^{*+} \left \right\rangle_S^2 = & \left\langle \Phi^{*+} \phi, \Phi^{*+} \phi \right\rangle = \langle \mathbf{O} \mathbf{K} \phi, \mathbf{O} \mathbf{K} \phi \rangle = \\ = (\mathbf{K} \phi, \mathbf{O}^* \mathbf{O} \mathbf{K} \phi) = & (\mathbf{K} \phi, \mathbf{K} \phi) = \ \mathbf{K} \phi\ _U^2. \end{aligned}$	$\begin{aligned} \ \Phi^{*+} \phi\ _S^2 = \langle \Phi^{*+} \phi, \Phi^{*+} \phi \rangle &= \langle \mathbf{O} \mathbf{K} \phi, \mathbf{O} \mathbf{K} \phi \rangle = \\ = (\mathbf{K} \phi, \mathbf{O}^* \mathbf{O} \mathbf{K} \phi) = & (\mathbf{K} \phi, \mathbf{K} \phi) = \ \mathbf{K} \phi\ _U^2. \end{aligned}$	$\begin{aligned} \ \Phi^{*+} \phi\ _S^2 = \langle \Phi^{*+} \phi, \Phi^{*+} \phi \rangle &= \langle \mathbf{O} \mathbf{K} \phi, \mathbf{O} \mathbf{K} \phi \rangle = \\ = (\mathbf{K} \phi, \mathbf{O}^* \mathbf{O} \mathbf{K} \phi) = & (\mathbf{K} \phi, \mathbf{K} \phi) = \ \mathbf{K} \phi\ _U^2. \end{aligned}$
penev_A_ EQ0108 (4.49)	022	■0.827	$a_{\mathbf{x}}(\mathbf{x}) = \frac{P_{\mathbf{x}_m}(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}.$	$a_{\mathbf{x}_m}(\mathbf{x}) = \frac{P_{\mathbf{x}_m}(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}.$	$a_{\mathbf{x}_m}(\mathbf{x}) = \frac{P_{\mathbf{x}_m}(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}.$
penev_A_EQ0066 (4.8)	018	■0.844	$K \, \Phi^* = O^*$	$K \Phi^* = O^*$	$\mathbf{K} \Phi^* = \mathbf{O}^*$
penev_A_ EQ0024 (2.27)	007	■0.852	$\mathbf{R} = \mathbf{R} \cdot \mathbf{P} = \sum_{\sigma>0} \sigma^2 \psi_{\sigma}(\psi_{\sigma},).$	$\mathbf{R} = \mathbf{R} \cdot \mathbf{P} = \sum_{\sigma>0} \sigma^2 \psi_{\sigma}(\psi_{\sigma},).$	$\mathbf{R} = \mathbf{R} \mathbf{P} = \sum_{\sigma>0} \sigma^2 \psi_{\sigma}(\psi_{\sigma},).$
penev_A_EQ0056 (3.7)	014	■0.852	$\mu_2 = \langle (\mathbf{u} - \mathbf{v})^2 \rangle = \langle \mathbf{u}^2 \rangle + \langle \mathbf{v}^2 \rangle - 2 \langle \mathbf{u} \mathbf{v} \rangle$	$\mu_2 = \langle (\mathbf{u} - \mathbf{v})^2 \rangle = \langle \mathbf{u}^2 \rangle + \langle \mathbf{v}^2 \rangle - 2 \langle \mathbf{u} \mathbf{v} \rangle$	$\mu_2 = \langle (\mathbf{u} - \mathbf{v})^2 \rangle = \langle \mathbf{u}^2 \rangle + \langle \mathbf{v}^2 \rangle - 2 \langle \mathbf{u} \mathbf{v} \rangle$
penev_A_ EQ0027 (2.30)	007	■0.857	$\Phi = \Phi \mathbf{P} = \Phi \sum_{\sigma>0} \psi_{\sigma}(\psi_{\sigma},) = \sum_{\sigma>0} \sigma \mathbf{s}_{\sigma}(\psi_{\sigma},).$	$\Phi = \Phi \mathbf{P} = \Phi \sum_{\sigma>0} \psi_{\sigma}(\psi_{\sigma},) = \sum_{\sigma>0} \sigma \mathbf{s}_{\sigma}(\psi_{\sigma},).$	$\Phi = \Phi \mathbf{P} = \Phi \sum_{\sigma>0} \psi_{\sigma}(\psi_{\sigma},) = \sum_{\sigma>0} \sigma \mathbf{s}_{\sigma}(\psi_{\sigma},).$
penev_A_EQ0065 (4.7)	018	■0.876	$\mathrm{K} \, \phi^t = \mathrm{K} \Phi^* \hat{t} \equiv \mathrm{O}^* \hat{t} \equiv \mathrm{O}^t.$	$\mathbf{K} \phi^t = \mathbf{K} \Phi^* \hat{t} \equiv \mathbf{O}^* \hat{t} \equiv \mathbf{O}^t.$	$\mathbf{K} \phi^t = \mathbf{K} \Phi^* \hat{t} \equiv \mathbf{O}^* \hat{t} \equiv \mathbf{O}^t.$
penev_A_ EQ0093 (4.34)	020	■0.881	$a_r(O) = (\psi_r, O).$	$a_r(O) = (\psi_r, O).$	$a_r(O) = (\psi_r, O).$
penev_A_ EQ0011 (2.12)	006	■0.881	$\langle \hat{t}, \Phi \hat{\mathbf{x}} \rangle = (\Phi^* \hat{t}, \hat{\mathbf{x}}) \equiv \phi^t(\mathbf{x}).$	$\langle \hat{t}, \Phi \hat{\mathbf{x}} \rangle = (\Phi^* \hat{t}, \hat{\mathbf{x}}) \equiv \phi^t(\mathbf{x}).$	$\langle \hat{t}, \Phi \hat{\mathbf{x}} \rangle = (\Phi^* \hat{t}, \hat{\mathbf{x}}) \equiv \phi^t(\mathbf{x}).$
penev_A_ EQ0080 (4.22)	018	■0.882	$\mathcal{P}[\phi] \propto \exp \left\{ -\frac{1}{2} \sum_{\mathbf{x}} O(\mathbf{x})^2 \right\} \propto \prod_{\mathbf{x}} \exp \left\{ -\frac{1}{2} O(\mathbf{x})^2 \right\}.$	$\mathcal{P}[\phi] \propto \exp \left\{ -\frac{1}{2} \sum_x O(\mathbf{x})^2 \right\} \propto \prod_{\mathbf{x}} \exp \left\{ -\frac{1}{2} O(\mathbf{x})^2 \right\}.$	$\mathcal{P}[\phi] \propto \exp \{ -\frac{1}{2} \sum_{\mathbf{x}} O(\mathbf{x})^2 \} \propto \prod_{\mathbf{x}} \exp \{ -\frac{1}{2} O(\mathbf{x})^2 \}.$
penev_A_ EQ0103 (4.44)	022	■0.887	$\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$	$\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$	$\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$
penev_A_ EQ0083 (4.25)	019	■0.891	$K_x(y) \equiv K \hat{x} = \sum_f \Psi_f \frac{1}{\sigma_f} (\Psi_f, \hat{x}) = \sum_f f e^{if(y-x)}$	$K_x(y) \equiv K \hat{x} = \sum_f \Psi_f \frac{1}{\sigma_f} (\Psi_f, \hat{x}) = \sum_f f e^{if(y-x)}$	$\mathbf{K}_{\mathbf{x}}(\mathbf{y}) \equiv \mathbf{K} \hat{\mathbf{x}} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \frac{1}{\sigma_{\mathbf{f}}} (\Psi_{\mathbf{f}}, \hat{\mathbf{x}}) = \sum_{\mathbf{f}} \mathbf{f} e^{i\mathbf{f}(\mathbf{y}-\mathbf{x})}$
penev_A_EQ0005 (2.5)	006	■0.894	$\begin{aligned} \mathbf{1}_U &= \frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}}(\hat{\mathbf{x}},)_U \\ \mathbf{1}_S &= \frac{1}{T} \sum_t \hat{t} \langle \hat{t}, \rangle_S. \end{aligned}$	$\begin{aligned} \mathbf{1}_U &= \frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}}(\hat{\mathbf{x}},)_U \\ \mathbf{1}_S &= \frac{1}{T} \sum_t \hat{t} \langle \hat{t}, \rangle_S. \end{aligned}$	$\begin{aligned} \mathbf{1}_U &= \frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}}(\hat{\mathbf{x}},)_U \\ \mathbf{1}_S &= \frac{1}{T} \sum_t \hat{t} \langle \hat{t}, \rangle_S. \end{aligned}$
penev_A_EQ0067 (4.9)	018	■0.896	$\langle \hat{t}, \mathbf{O} \hat{\mathbf{x}} \rangle = (\mathbf{O}^* \hat{t}, \hat{\mathbf{x}}) \equiv O^t(\mathbf{x}).$	$\langle \hat{t}, \mathbf{O} \hat{\mathbf{x}} \rangle = (\mathbf{O}^* \hat{t}, \hat{\mathbf{x}}) \equiv O^t(\mathbf{x}).$	$\langle \hat{t}, \mathbf{O} \hat{\mathbf{x}} \rangle = (\mathbf{O}^* \hat{t}, \hat{\mathbf{x}}) \equiv O^t(\mathbf{x}).$
penev_A_ EQ0114 (4.55)	022	■0.899	$\mathbf{s}_{ } = \sum_{m=1}^M O_m \mathbf{p}_m$	$\mathbf{s}_{ } = \sum_{m=1}^M O_m \mathbf{p}_m$	$\mathbf{s}_{ } = \sum_{m=1}^M O_m \mathbf{p}_m$
penev_A_ EQ0086 (4.28)	019	■0.908	$\mathbf{O}^* \mathbf{O} = \mathbf{P}$	$\mathbf{O}^* \mathbf{O} = \mathbf{P}$	$\mathbf{O}^* \mathbf{O} = \mathbf{P}$
penev_A_ EQ0072 (4.14)	018	■0.911	$\mathbf{O} \Psi_{\mathbf{f}} = \mathbf{o}_{\mathbf{f}} \in S.$	$\mathbf{O} \Psi_{\mathbf{f}} = \mathbf{o}_{\mathbf{f}} \in S.$	$\mathbf{O} \Psi_{\mathbf{f}} = \mathbf{o}_{\mathbf{f}} \in S.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

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penev_A_EQ0051 (3.2)	014	■ 0.914	$T \times T = \{(p, p)\}_{p \in T} \cup C_{\text{same}} \cup C_{\text{diff}}.$	$T \times T = \{(p, p)\}_{p \in T} \cup C_{\text{same}} \cup C_{\text{diff}}.$	$T \times T = \{(p, p)\}_{p \in T} \cup C_{\text{same}} \cup C_{\text{diff}}.$
penev_A_EQ0090 (4.31)	020	■ 0.925	$0=\mathrm{K} \ \ \mathrm{\phi}=\mathrm{0}^{\ast} \ \ \mathrm{\Phi}^{\ast+} \ \ \mathrm{\phi}=\sum_{r=1}^N \mathrm{a}_{\mathrm{r}} \ \ \mathrm{\psi}_{\mathrm{r}}$	$O = \mathbf{K}\phi = \mathbf{O}^*\Phi^{*+}\phi = \sum_{r=1}^N a_r\psi_r$	$O = \mathbf{K}\phi = \mathbf{O}^*\Phi^{*+}\phi = \sum_{r=1}^N a_r\psi_r$
penev_A_EQ0085 (4.27)	019	■ 0.928	$\mathrm{K}^{\mathrm{n}}=\sum_{\mathrm{f}}\mathrm{\Psi}_{\mathrm{f}}\left(\frac{1}{\sigma_{\mathrm{f}}}\right)^{\mathrm{n}}(\Psi_{\mathrm{f}},).$	$K^n = \sum_{\mathrm{f}} \Psi_{\mathrm{f}} \left(\frac{1}{\sigma_{\mathrm{f}}} \right)^n (\Psi_{\mathrm{f}},).$	$\mathbf{K}^n = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \left(\frac{1}{\sigma_{\mathbf{f}}} \right)^n (\Psi_{\mathbf{f}},).$
penev_A_EQ0097 (4.38)	021	■ 0.929	$\mathrm{q}_{\mathrm{x}_m} \equiv \mathbf{O} \hat{\mathbf{x}}_m.$	$\mathbf{q}_{\mathbf{x}_m} \equiv \mathbf{O} \hat{\mathbf{x}}_m.$	$\mathbf{q}_{\mathbf{x}_m} \equiv \mathbf{O} \hat{\mathbf{x}}_m.$
penev_A_EQ0032 (2.35)	007	■ 0.932	$\mathrm{cal}\left[\mathrm{\Phi}^{\ast}\mathrm{s}\right]\propto\exp\left[-\frac{1}{2}\left\ \mathbf{P}'\mathbf{s}\right\ _S^2\right].$	$\mathcal{P}\left[\Phi^*\mathbf{s}\right]\propto\exp\left\{-\frac{1}{2}\left\ \mathbf{P}'\mathbf{s}\right\ _S^2\right\}.$	$\mathcal{P}[\Phi^*\mathbf{s}]\propto\exp\left\{-\frac{1}{2}\left\ \mathbf{P}'\mathbf{s}\right\ _S^2\right\}.$
penev_A_EQ0042 (2.45)	008	■ 0.934	$\begin{aligned} \Phi_N &= \sum_{r=1}^N \sigma_r \mathbf{s}_r(\psi_r,) \\ (\Phi_N)^* \Phi_N &= \mathbf{R}_N = \sum_{r=1}^N \sigma_r^2 (\psi_r,) \\ \mathrm{tr} \mathbf{R}_N &= \sum_{r=1}^N \sigma_r^2. \end{aligned}$	$\begin{aligned} \Phi_N &= \sum_{r=1}^N \sigma_r \mathbf{s}_r(\psi_r,) \\ (\Phi_N)^* \Phi_N &= \mathbf{R}_N = \sum_{r=1}^N \sigma_r^2 (\psi_r,) \\ \mathrm{tr} \mathbf{R}_N &= \sum_{r=1}^N \sigma_r^2. \end{aligned}$	$\begin{aligned} \Phi_N &= \sum_{r=1}^N \sigma_r \mathbf{s}_r(\psi_r,) \\ (\Phi_N)^* \Phi_N &= \mathbf{R}_N = \sum_{r=1}^N \sigma_r^2 (\psi_r,) \\ \mathrm{tr} \mathbf{R}_N &= \sum_{r=1}^N \sigma_r^2. \end{aligned}$
penev_A_EQ0101 (4.42)	022	■ 0.945	$\mathrm{s}_{\parallel}=\mathrm{Q}_{\mathbf{x}_m}\mathbf{s}=\mathbf{p}_{\mathbf{x}_m}\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\right\rangle=O_m\mathbf{p}_{\mathbf{x}_m}$	$\mathbf{s}_{\parallel} = \mathbf{Q}_{\mathbf{x}_m} \mathbf{s} = \mathbf{p}_{\mathbf{x}_m} \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s} \rangle = O_m \mathbf{p}_{\mathbf{x}_m}$	$\mathbf{s}_{\parallel} = \mathbf{Q}_{\mathbf{x}_m} \mathbf{s} = \mathbf{p}_{\mathbf{x}_m} \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s} \rangle = O_m \mathbf{p}_{\mathbf{x}_m}$
penev_A_EQ0099 (4.40)	022	■ 0.948	$\mathrm{p}_{x_m}=\frac{\mathbf{q}_{x_m}}{\left\langle\mathbf{q}_{x_m},\mathbf{q}_{x_m}\right\rangle}.$	$\mathbf{p}_{x_m} = \frac{\mathbf{q}_{x_m}}{\langle \mathbf{q}_{x_m}, \mathbf{q}_{x_m} \rangle}.$	$\mathbf{p}_{\mathbf{x}_m} = \frac{\mathbf{q}_{\mathbf{x}_m}}{\langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{q}_{\mathbf{x}_m} \rangle}.$
penev_A_EQ0115 (4.56)	022	■ 0.949	$\left\langle\mathbf{q}_m,\mathbf{s}_{\parallel}\right\rangle=\sum_{l=1}^{ \mathcal{M} }O_l\left\langle\mathbf{q}_m,\mathbf{p}_l\right\rangle=\sum_{l=1}^{ \mathcal{M} }O_l\delta_{ml}=O_m=\left\langle\mathbf{q}_m,\mathbf{s}\right\rangle$	$\langle \mathbf{q}_m, \mathbf{s}_{\parallel} \rangle = \sum_{l=1}^{ \mathcal{M} } O_l \langle \mathbf{q}_m, \mathbf{p}_l \rangle = \sum_{l=1}^{ \mathcal{M} } O_l \delta_{ml} = O_m = \langle \mathbf{q}_m, \mathbf{s} \rangle$	$\langle \mathbf{q}_m, \mathbf{s}_{\parallel} \rangle = \sum_{l=1}^{ \mathcal{M} } O_l \langle \mathbf{q}_m, \mathbf{p}_l \rangle = \sum_{l=1}^{ \mathcal{M} } O_l \delta_{ml} = O_m = \langle \mathbf{q}_m, \mathbf{s} \rangle$
penev_A_EQ0098 (4.39)	022	■ 0.960	$\mathrm{Q}_{x_m}=\frac{\mathbf{q}_{x_m}\left\langle\mathbf{q}_{x_m},\right\rangle}{\left\langle\mathbf{q}_{x_m},\mathbf{q}_{x_m}\right\rangle}=\mathbf{p}_{x_m}\left\langle\mathbf{q}_{x_m},\right\rangle$	$\mathbf{Q}_{x_m} = \frac{\mathbf{q}_{x_m} \langle \mathbf{q}_{x_m}, \rangle}{\langle \mathbf{q}_{x_m}, \mathbf{q}_{x_m} \rangle} = \mathbf{p}_{x_m} \langle \mathbf{q}_{x_m}, \rangle$	$\mathbf{Q}_{\mathbf{x}_m} = \frac{\mathbf{q}_{\mathbf{x}_m} \langle \mathbf{q}_{\mathbf{x}_m}, \rangle}{\langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{q}_{\mathbf{x}_m} \rangle} = \mathbf{p}_{\mathbf{x}_m} \langle \mathbf{q}_{\mathbf{x}_m}, \rangle$
penev_A_EQ0095 (4.36)	021	■ 0.968	$O(\mathbf{x})=\sum_{m=1}^{\left M\right }O_ma_m(\mathbf{x})$	$O(\mathbf{x}) = \sum_{m=1}^{ M } O_m a_m(\mathbf{x})$	$O(\mathbf{x}) = \sum_{m=1}^{ M } O_m a_m(\mathbf{x})$
penev_A_EQ0049 (2.52)	008	■ 0.975	$\left\ \phi_N^{rec}\right\ _U^2=\left(\phi_N^{rec},\phi_N^{rec}\right)=\sum_{r=1}^N\sigma^2a_r^2$	$\ \phi_N^{rec}\ _U^2 = (\phi_N^{rec}, \phi_N^{rec}) = \sum_{r=1}^N \sigma^2 a_r^2$	$\ \phi_N^{rec}\ _U^2 = (\phi_N^{rec}, \phi_N^{rec}) = \sum_{r=1}^N \sigma^2 a_r^2$
penev_A_EQ0044 (2.47)	008	■ 0.976	$a_r(\phi)\equiv\sigma_r^{-1}(\psi_r,\phi)$	$a_r(\phi) \equiv \sigma_r^{-1}(\psi_r, \phi)$	$a_r(\phi) \equiv \sigma_r^{-1}(\psi_r, \phi)$
penev_A_EQ0078 (4.20)	018	■ 0.977	$\mathrm{OK}=\mathrm{O}\mathrm{O}^*\mathrm{\Phi}^{*+}=\mathrm{P}'\mathrm{\Phi}^{*+}=\mathrm{\Phi}^{*+}$	$\mathrm{OK} = \mathrm{O}\mathrm{O}^*\Phi^{*+} = \mathrm{P}'\Phi^{*+} = \Phi^{*+}$	$\mathrm{OK} = \mathrm{O}\mathrm{O}^*\Phi^{*+} = \mathrm{P}'\Phi^{*+} = \Phi^{*+}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value					

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penev_A_EQ00008 (2.9)	006	■ 0.977	$\begin{aligned} \ \phi\ _U^2 &= (\phi, \phi) = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi(\mathbf{x}) \\ \ \mathbf{s}\ _S^2 &= \langle \mathbf{s}, \mathbf{s} \rangle = \frac{1}{T} \sum_t s(t) s(t) \end{aligned}$	$\ \phi\ _U^2 = (\phi, \phi) = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi(\mathbf{x})$ $\ \mathbf{s}\ _S^2 = \langle \mathbf{s}, \mathbf{s} \rangle = \frac{1}{T} \sum_t s(t) s(t)$	$\ \phi\ _U^2 = (\phi, \phi) = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi(\mathbf{x})$ $\ \mathbf{s}\ _S^2 = \langle \mathbf{s}, \mathbf{s} \rangle = \frac{1}{T} \sum_t s(t) s(t).$
penev_A_EQ00088 (4.29)	020	■ 0.980	$\mathbf{K}^{(-2)} = \mathbf{R} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-2} (\psi_r,)$	$\mathbf{K}^{(-2)} = \mathbf{R} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-2} (\psi_r,)$	$\mathbf{K}^{(-2)} = \mathbf{R} = \sum_{r=1}^N \psi_r \left(\frac{1}{\sigma_r} \right)^{-2} (\psi_r,)$
penev_A_EQ00081 (4.23)	018	■ 0.981	$\mathrm{O}^{*+} \mathrm{O} = \mathrm{O} \mathrm{O} = \mathrm{O} \mathrm{K} \phi = \Phi^{*+} \phi.$	$\mathrm{O}^{*+} \mathrm{O} = \mathrm{O} \mathrm{O} = \mathrm{O} \mathrm{K} \phi = \Phi^{*+} \phi.$	$\mathrm{O}^{*+} \mathrm{O} = \mathrm{O} \mathrm{O} = \mathrm{O} \mathrm{K} \phi = \Phi^{*+} \phi.$
penev_A_EQ00076 (4.18)	018	■ 0.981	$0 \ 0^{\{*\}} = \sum_f s_{\{f\}} \left(s_{\{f\}}, \right) = P^{\{\prime\}}$	$O O^* = \sum_f s_f (s_f,) = P'.$	$O O^* = \sum_{\mathbf{f}} \mathbf{s}_{\mathbf{f}} (\mathbf{s}_{\mathbf{f}},) = \mathbf{P}'.$
penev_A_EQ00001 (2.1)	006	■ 0.982	$R(\mathbf{x}, \mathbf{y}) = \langle \phi^t(\mathbf{x}) \phi^t(\mathbf{y}) \rangle_T$	$R(\mathbf{x}, \mathbf{y}) = \langle \phi^t(\mathbf{x}) \phi^t(\mathbf{y}) \rangle_T$	$R(\mathbf{x}, \mathbf{y}) = \langle \phi^t(\mathbf{x}) \phi^t(\mathbf{y}) \rangle_T$
penev_A_EQ00020 (2.23)	007	■ 0.986	$\ker \Phi^* \Phi = \ker \Phi.$	$\ker \Phi^* \Phi = \ker \Phi.$	$\ker \Phi^* \Phi = \ker \Phi.$
penev_A_EQ00052 (3.3)	014	■ 0.991	$\mathrm{d}^{(p,q)} = (\Phi_N)^{*+} \phi^p \quad (\Phi_N)^{*+} \phi^q \in S$	$\mathbf{d}^{(p,q)} = (\Phi_N)^{*+} \phi^p \quad (\Phi_N)^{*+} \phi^q \in S$	$\mathbf{d}^{(p,q)} = (\Phi_N)^{*+} \phi^p \quad (\Phi_N)^{*+} \phi^q \in S$
penev_A_EQ00009 (2.10)	006	■ 0.995	$(\phi, \phi') = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi'(\mathbf{x})$	$(\phi, \phi') = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi'(\mathbf{x})$	$(\phi, \phi') = \frac{1}{V} \sum_{\mathbf{x}} \phi(\mathbf{x}) \phi'(\mathbf{x})$
penev_A_EQ00050 (3.1)	013	■ 0.995	$\epsilon = \ \mathbf{P}_N \phi^k - \mathbf{P}_N \phi\ _U$	$\epsilon = \ \mathbf{P}_N \phi^k - \mathbf{P}_N \phi\ _U$	$\epsilon = \ \mathbf{P}_N \phi^k - \mathbf{P}_N \phi\ _U$
penev_A_EQ00043 (2.46)	008	■ 0.995	$(\Phi_N)^{*+} = \sum_{r=1}^N \sigma_r^{-1} \mathbf{s}_r (\psi_r,)$	$(\Phi_N)^{*+} = \sum_{r=1}^N \sigma_r^{-1} \mathbf{s}_r (\psi_r,)$	$(\Phi_N)^{*+} = \sum_{r=1}^N \sigma_r^{-1} \mathbf{s}_r (\psi_r,).$
penev_A_EQ00025 (2.28)	007	■ 0.997	$\sigma \mathbf{s}_\sigma \equiv \Phi \psi_\sigma$	$\sigma \mathbf{s}_\sigma \equiv \Phi \psi_\sigma$	$\sigma \mathbf{s}_\sigma \equiv \Phi \psi_\sigma$
penev_A_EQ00037 (2.40)	008	■ 0.997	$(\phi^t, \phi^t) = \sum_{\sigma>0} \sigma^2 a_\sigma^{t^2}.$	$(\phi^t, \phi^t) = \sum_{\sigma>0} \sigma^2 a_\sigma^{t^2}.$	$(\phi^t, \phi^t) = \sum_{\sigma>0} \sigma^2 a_\sigma^{t^2}.$
penev_A_EQ00048 (2.51)	008	■ 0.997	$\phi^{err} \equiv \phi - \phi^{rec} = (1 - \mathbf{P}_N) \phi$	$\phi^{err} \equiv \phi - \phi^{rec} = (1 - \mathbf{P}_N) \phi$	$\phi^{err} \equiv \phi - \phi^{rec} = (1 - \mathbf{P}_N) \phi$
penev_A_EQ00033 (2.36)	007	■ 0.997	$\ \mathbf{P}' \mathbf{s}\ _S^2 = \langle \mathbf{P}' \mathbf{s}, \mathbf{P}' \mathbf{s} \rangle = \sum_{\sigma>0} (\mathbf{s}_\sigma, \mathbf{s})^2.$	$\ \mathbf{P}' \mathbf{s}\ _S^2 = \langle \mathbf{P}' \mathbf{s}, \mathbf{P}' \mathbf{s} \rangle = \sum_{\sigma>0} (\mathbf{s}_\sigma, \mathbf{s})^2.$	$\ \mathbf{P}' \mathbf{s}\ _S^2 = \langle \mathbf{P}' \mathbf{s}, \mathbf{P}' \mathbf{s} \rangle = \sum_{\sigma>0} (\mathbf{s}_\sigma, \mathbf{s})^2.$
penev_A_EQ00038 (2.41)	008	■ 0.998	$(\psi_\sigma, \mathbf{R} \psi_{\sigma'}) = \langle \Phi \psi_\sigma, \Phi \psi_{\sigma'} \rangle = \sigma \sigma' \frac{1}{T} \sum_t a_\sigma^t a_{\sigma'}^t,$	$(\psi_\sigma, \mathbf{R} \psi_{\sigma'}) = \langle \Phi \psi_\sigma, \Phi \psi_{\sigma'} \rangle = \sigma \sigma' \frac{1}{T} \sum_t a_\sigma^t a_{\sigma'}^t,$	$\langle \psi_\sigma, \mathbf{R} \psi_{\sigma'} \rangle = \langle \Phi \psi_\sigma, \Phi \psi_{\sigma'} \rangle = \sigma \sigma' \frac{1}{T} \sum_t a_\sigma^t a_{\sigma'}^t,$
penev_A_EQ00082 (4.24)	019	■ 0.998	$\mathbf{K}^{-1} = \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \sigma_{\mathbf{f}} (\Psi_{\mathbf{f}},).$	$\mathbf{K}^{-1} = \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \sigma_{\mathbf{f}} (\Psi_{\mathbf{f}},).$	$\mathbf{K}^{-1} = \Phi^* \mathbf{O}^{*+} = \sum_{\mathbf{f}} \Psi_{\mathbf{f}} \sigma_{\mathbf{f}} (\Psi_{\mathbf{f}},).$
penev_A_EQ00029 (2.32)	007	■ 0.998	$\Phi^{*+} = \sum_{\sigma>0} \sigma^{-1} \mathbf{s}_\sigma (\psi_\sigma,).$	$\Phi^{*+} = \sum_{\sigma>0} \sigma^{-1} \mathbf{s}_\sigma (\psi_\sigma,).$	$\Phi^{*+} = \sum_{\sigma>0} \sigma^{-1} \mathbf{s}_\sigma (\psi_\sigma,).$
penev_A_EQ00022 (2.25)	007	■ 0.998	$\mathbf{R} = \sum_{\sigma} \sigma^2 \psi_\sigma (\psi_\sigma,)$	$\mathbf{R} = \sum_{\sigma} \sigma^2 \psi_\sigma (\psi_\sigma,)$	$\mathbf{R} = \sum_{\sigma} \sigma^2 \psi_\sigma (\psi_\sigma,)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value					

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penev_A_EQ0062 (4.4)	017	■ 0.999	$\sigma_{\mathbf{f}}^2 \equiv \tilde{r}(\mathbf{f}) \propto \frac{1}{ \mathbf{f} ^\alpha}$ $\propto \frac{1}{ \mathbf{f} ^\alpha}$	$\sigma_{\mathbf{f}}^2 \equiv \tilde{r}(\mathbf{f}) \propto \frac{1}{ \mathbf{f} ^\alpha}$	$\sigma_{\mathbf{f}}^2 \equiv \tilde{r}(\mathbf{f}) \propto \frac{1}{ \mathbf{f} ^\alpha}$
penev_A_EQ0045 (2.38)	008	■ 0.999	$\left(\Phi_N\right)^{++} \phi=\sum_{r=1}^N \sigma_r^{-1} \mathbf{s}_r\left(\psi_r, \phi\right)=\sum_{r=1}^N a_r \mathbf{s}_r .$ $\sigma_{\mathbf{r}}^{-1} \mathbf{s}_r\left(\psi_r, \phi\right)=\sum_{r=1}^N a_r \mathbf{s}_r .$	$(\Phi_N)^{++} \phi = \sum_{r=1}^N \sigma_r^{-1} \mathbf{s}_r(\psi_r, \phi) = \sum_{r=1}^N a_r \mathbf{s}_r.$	$(\Phi_N)^{++} \phi = \sum_{r=1}^N \sigma_r^{-1} \mathbf{s}_r(\psi_r, \phi) = \sum_{r=1}^N a_r \mathbf{s}_r.$
penev_A_EQ0012 (2.13)	006	■ 0.999	$\Phi \hat{\mathbf{x}}=\frac{1}{T} \sum_t \hat{t} \phi^t(\mathbf{x}) .$ $\phi^t(\mathbf{x})$	$\Phi \hat{\mathbf{x}} = \frac{1}{T} \sum_t \hat{t} \phi^t(\mathbf{x}).$	$\Phi \hat{\mathbf{x}} = \frac{1}{T} \sum_t \hat{t} \phi^t(\mathbf{x}).$
penev_A_EQ0091 (4.32)	020	■ 0.999	$O(\mathbf{x})=\left(\hat{\mathbf{x}}, O\right)=\sum_{r=1}^N a_r \Psi_r(\mathbf{x})$ $\Psi_r(\mathbf{x})$	$O(\mathbf{x}) = (\hat{\mathbf{x}}, O) = \sum_{r=1}^N a_r \Psi_r(\mathbf{x})$	$O(\mathbf{x}) = (\hat{\mathbf{x}}, O) = \sum_{r=1}^N a_r \Psi_r(\mathbf{x}).$
penev_A_EQ0007 (2.7)	006	■ 1.000	$\mathrm{s}=\frac{1}{T} \sum_t \hat{s}(t) \in S .$ $\hat{s}(t)$	$\mathbf{s} = \frac{1}{T} \sum_t \hat{s}(t) \in S.$	$\mathbf{s} = \frac{1}{T} \sum_t \hat{s}(t) \in S.$
penev_A_EQ0023 (2.26)	007	■ 1.000	$\mathbf{P}=\sum_{\sigma>0} \psi_{\sigma}\left(\psi_{\sigma},\right) .$ ψ_{σ}	$\mathbf{P} = \sum_{\sigma>0} \psi_{\sigma}(\psi_{\sigma},).$	$\mathbf{P} = \sum_{\sigma>0} \psi_{\sigma}(\psi_{\sigma},).$
penev_A_EQ0105	022	■ 1.000	$O^{\{r \in \mathcal{C}\}}(\mathbf{x})=O_m\left(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{p}_{\mathbf{x}_m}\right)=\frac{O_m}{\left\langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{q}_{\mathbf{x}_m}\right\rangle}\left(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}\right) .$ $\frac{O_m}{\left\langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{q}_{\mathbf{x}_m}\right\rangle}\left(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}\right) .$	$O^{rec}(\mathbf{x}) = O_m(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{p}_{\mathbf{x}_m}) = \frac{O_m}{\langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{q}_{\mathbf{x}_m} \rangle}(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m})$	$O^{rec}(\mathbf{x}) = O_m(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{p}_{\mathbf{x}_m}) = \frac{O_m}{\langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{q}_{\mathbf{x}_m} \rangle}(\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{q}_{\mathbf{x}_m}).$
penev_A_EQ0110 (4.51)	022	■ 1.000	$\phi^{\{r \in \mathcal{C}\}}(\mathbf{x})=O_m \frac{K^{(-1)} \mathbf{x}_m(\mathbf{x})}{P_{\mathbf{x}_m}\left(\mathbf{x}_m\right)}$ $\frac{K^{(-1)} \mathbf{x}_m(\mathbf{x})}{P_{\mathbf{x}_m}\left(\mathbf{x}_m\right)}$	$\phi^{rec}(\mathbf{x}) = O_m \frac{K^{(-1)} \mathbf{x}_m(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}$	$\phi^{rec}(\mathbf{x}) = O_m \frac{K^{(-1)} \mathbf{x}_m(\mathbf{x})}{P_{\mathbf{x}_m}(\mathbf{x}_m)}$
penev_A_EQ0002 (2.2)	006	■ 1.000	$\mathcal{P}[\phi(\mathbf{x})] \propto \exp \left\{-\frac{1}{2} \int_V d \mathbf{x} \phi(\mathbf{x}) \int_V d \mathbf{y} R^{-1}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{y})\right\}$ $\exp \left\{-\frac{1}{2} \int_V d \mathbf{x} \phi(\mathbf{x}) \int_V d \mathbf{y} R^{-1}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{y})\right\}$	$\mathcal{P}[\phi(\mathbf{x})] \propto \exp \left\{-\frac{1}{2} \int_V d \mathbf{x} \phi(\mathbf{x}) \int_V d \mathbf{y} R^{-1}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{y})\right\}$	$\mathcal{P}[\phi(\mathbf{x})] \propto \exp \left\{-\frac{1}{2} \int_V d \mathbf{x} \phi(\mathbf{x}) \int_V d \mathbf{y} R^{-1}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{y})\right\}.$
penev_A_EQ0003 (2.3)	006	■ 1.000	$\Phi: U \mapsto S$ $U \mapsto S$	$\Phi: U \mapsto S$	$\Phi: U \mapsto S$
penev_A_EQ0006 (2.6)	006	■ 1.000	$\phi=\frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}} \phi(\mathbf{x}) \in U$ $\hat{\mathbf{x}} \phi(\mathbf{x}) \in U$	$\phi = \frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}} \phi(\mathbf{x}) \in U$	$\phi = \frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}} \phi(\mathbf{x}) \in U$
penev_A_EQ0010 (2.11)	006	■ 1.000	$\langle \mathbf{s}, \Phi \mathbf{u}\rangle=\left\langle \Phi^* \mathbf{s}, \mathbf{u}\right\rangle .$ $\left\langle \Phi^* \mathbf{s}, \mathbf{u}\right\rangle .$	$\langle \mathbf{s}, \Phi \mathbf{u} \rangle = (\Phi^* \mathbf{s}, \mathbf{u}).$	$\langle \mathbf{s}, \Phi \mathbf{u} \rangle = (\Phi^* \mathbf{s}, \mathbf{u}).$
penev_A_EQ0013 (2.14)	006	■ 1.000	$\langle \Phi \hat{\mathbf{x}}, \Phi \hat{\mathbf{y}}\rangle=\frac{1}{T} \sum_t \phi^t(\mathbf{x}) \phi^t(\mathbf{y})$ $\phi^t(\mathbf{x}) \phi^t(\mathbf{y})$	$\langle \Phi \hat{\mathbf{x}}, \Phi \hat{\mathbf{y}} \rangle = \frac{1}{T} \sum_t \phi^t(\mathbf{x}) \phi^t(\mathbf{y})$	$\langle \Phi \hat{\mathbf{x}}, \Phi \hat{\mathbf{y}} \rangle = \frac{1}{T} \sum_t \phi^t(\mathbf{x}) \phi^t(\mathbf{y})$
penev_A_EQ0014 (2.15)	006	■ 1.000	$\Phi^* \hat{t} \equiv \mathbf{1}_U \Phi^* \hat{t}=\frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}} \phi^t(\mathbf{x})=\phi^t .$ $\frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}} \phi^t(\mathbf{x}) = \phi^t.$	$\Phi^* \hat{t} \equiv \mathbf{1}_U \Phi^* \hat{t} = \frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}} \phi^t(\mathbf{x}) = \phi^t.$	$\Phi^* \hat{t} \equiv \mathbf{1}_U \Phi^* \hat{t} = \frac{1}{V} \sum_{\mathbf{x}} \hat{\mathbf{x}} \phi^t(\mathbf{x}) = \phi^t.$
penev_A_EQ0015 (2.16)	007	■ 1.000	$\mathcal{P}\left[\Phi^* \mathbf{s}\right] \propto \exp \left\{-\frac{1}{2}\left\ \mathbf{s}\right\ _S^2\right\}$ $\exp \left\{-\frac{1}{2}\left\ \mathbf{s}\right\ _S^2\right\}$	$\mathcal{P}\left[\Phi^* \mathbf{s}\right] \propto \exp \left\{-\frac{1}{2}\left\ \mathbf{s}\right\ _S^2\right\}$	$\mathcal{P}\left[\Phi^* \mathbf{s}\right] \propto \exp \left\{-\frac{1}{2}\left\ \mathbf{s}\right\ _S^2\right\}$
penev_A_EQ0016 (2.17)	007	■ 1.000	$\Phi^{*+}: U \mapsto S$ $U \mapsto S$	$\Phi^{*+}: U \mapsto S$	$\Phi^{*+}: U \mapsto S$
penev_A_EQ0017 (2.18)	007	■ 1.000	$\Phi^* \Phi^{*+} \Phi^* \mathbf{s}=\Phi^* \mathbf{s} .$ $\Phi^* \mathbf{s} .$	$\Phi^* \Phi^{*+} \Phi^* \mathbf{s} = \Phi^* \mathbf{s}.$	$\Phi^* \Phi^{*+} \Phi^* \mathbf{s} = \Phi^* \mathbf{s}.$
penev_A_EQ0019 (2.22)	007	■ 1.000	$\left(\mathbf{u}, \Phi^* \Phi \mathbf{u}\right)=\left\langle \Phi \mathbf{u}, \Phi \mathbf{u}\right\rangle=\left\ \Phi \mathbf{u}\right\ _S^2$ $\left\ \Phi \mathbf{u}\right\ _S^2$	$(\mathbf{u}, \Phi^* \Phi \mathbf{u}) = \langle \Phi \mathbf{u}, \Phi \mathbf{u} \rangle = \ \Phi \mathbf{u}\ _S^2$	$(\mathbf{u}, \Phi^* \Phi \mathbf{u}) = \langle \Phi \mathbf{u}, \Phi \mathbf{u} \rangle = \ \Phi \mathbf{u}\ _S^2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
penev_A_ E00036 (2.39)	008	■ 1.000	$\phi^t=\sum_{\sigma>0}\sigma a_{\sigma}^t\psi_{\sigma}\cdot$	$\phi^t=\sum_{\sigma>0}\sigma a_{\sigma}^t\psi_{\sigma}.$	$\phi^t=\sum_{\sigma>0}\sigma a_{\sigma}^t\psi_{\sigma}.$
penev_A_ E00046 (2.49)	008	■ 1.000	$\mathcal{P}[\phi]\propto\exp\left\{-\frac{1}{2}\sum_{r=1}^Na_r^2\right\}\propto\prod_{r=1}^N\exp\left\{-\frac{1}{2}a_r^2\right\}$	$\mathcal{P}[\phi]\propto\exp\left\{-\frac{1}{2}\sum_{r=1}^Na_r^2\right\}\propto\prod_{r=1}^N\exp\left\{-\frac{1}{2}a_r^2\right\}$	$\mathcal{P}[\phi]\propto\exp\{-\frac{1}{2}\sum_{r=1}^Na_r^2\}\propto\prod_{r=1}^N\exp\{-\frac{1}{2}a_r^2\}$
penev_A_E00053 (3.4)	014	■ 1.000	$\mu\left(\mathrm{d}^{(p,q)}\right)=\sum_{r=1}^Nw_r\left(a_r^p-a_r^q\right)^2=\sum_{r=1}^Nw_rD_r^{(p,q)}\equiv D^{(p,q)}$	$\mu\left(\mathrm{d}^{(p,q)}\right)=\sum_{r=1}^Nw_r\left(a_r^p-a_r^q\right)^2=\sum_{r=1}^Nw_rD_r^{(p,q)}\equiv D^{(p,q)}$	$\mu(\mathrm{d}^{(p,q)})=\sum_{r=1}^Nw_r(a_r^p-a_r^q)^2=\sum_{r=1}^Nw_rD_r^{(p,q)}\equiv D^{(p,q)}$
penev_A_E00054 (3.5)	014	■ 1.000	$D_{\{r\}^{\{p,q\}}}\equiv\left(a_{\{r\}^{\{p\}}}-a_{\{r\}^{\{q\}}}\right)^2.$	$D_r^{(p,q)}\equiv\left(a_r^p-a_r^q\right)^2.$	$D_r^{(p,q)}\equiv\left(a_r^p-a_r^q\right)^2.$
penev_A_E00058 (3.9)	015	■ 1.000	$w_r^{\text{naive}}\propto\left(m_r^{\text{same}}-m_r^{\text{diff}}\right)\left(H^{\text{intra}}\right)^{-1}_{rr}.$	$w_r^{\text{naive}}\propto\left(m_r^{\text{same}}-m_r^{\text{diff}}\right)\left(H^{\text{intra}}\right)^{-1}_{rr}.$	$w_r^{\text{naive}}\propto\left(m_r^{\text{same}}-m_r^{\text{diff}}\right)\left(H^{\text{intra}}\right)^{-1}_{rr}.$
penev_A_E00059 (4.1)	017	■ 1.000	$R(\mathbf{x},\mathbf{y})=r(\mathbf{x}-\mathbf{y}).$	$R(\mathbf{x},\mathbf{y})=r(\mathbf{x}-\mathbf{y}).$	$R(\mathbf{x},\mathbf{y})=r(\mathbf{x}-\mathbf{y}).$
penev_A_E00060 (4.2)	017	■ 1.000	$R(\mathbf{x},\mathbf{y})=r(\mathbf{x}-\mathbf{y})=\rho(\ell).$	$R(\mathbf{x},\mathbf{y})=r(\mathbf{x}-\mathbf{y})=\rho(\ell).$	$R(\mathbf{x},\mathbf{y})=\tau(\mathbf{x}-\mathbf{y})=\rho(\ell).$
penev_A_E00064 (4.6)	018	■ 1.000	$\mathrm{K}\colon U\rightarrow U$	$\mathbf{K}\colon U\rightarrow U$	$\mathbf{K}\colon U\rightarrow U$
penev_A_ E00068 (4.10)	018	■ 1.000	$\langle O(\mathbf{x})O(\mathbf{y})\rangle_T=\frac{1}{T}\sum_tO^t(\mathbf{x})O^t(\mathbf{y})\propto\delta_{\mathbf{x},\mathbf{y}}$	$\langle O(\mathbf{x})O(\mathbf{y})\rangle_T=\frac{1}{T}\sum_tO^t(\mathbf{x})O^t(\mathbf{y})\propto\delta_{\mathbf{x},\mathbf{y}}$	$\langle O(\mathbf{x})O(\mathbf{y})\rangle_T=\frac{1}{T}\sum_tO^t(\mathbf{x})O^t(\mathbf{y})\propto\delta_{\mathbf{x},\mathbf{y}}$
penev_A_ E00089 (4.30)	020	■ 1.000	$\mathbf{K}^{(n)}=\sum_{r=1}^N\psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right).$	$\mathbf{K}^{(n)}=\sum_{r=1}^N\psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right).$	$\mathbf{K}^{(n)}=\sum_{r=1}^N\psi_r\left(\frac{1}{\sigma_r}\right)^n\left(\psi_r,\right).$
penev_A_ E00102 (4.43)	022	■ 1.000	$\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\perp}\right\rangle=\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}-\mathbf{s}_{\parallel}\right\rangle=\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\right\rangle-\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\parallel}\right\rangle=O_m-O_m=0.$	$\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\perp}\right\rangle=\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}-\mathbf{s}_{\parallel}\right\rangle=\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\right\rangle-\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\parallel}\right\rangle=O_m-O_m=0.$	$\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\perp}\right\rangle=\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}-\mathbf{s}_{\parallel}\right\rangle=\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}\right\rangle-\left\langle\mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\parallel}\right\rangle=O_m-O_m=0.$
penev_A_ E00109 (4.50)	022	■ 1.000	$\phi^{rec}(\mathbf{x})=(\hat{\mathbf{x}},\Phi^*\mathbf{s}^{rec})$	$\phi^{rec}(\mathbf{x})=(\hat{\mathbf{x}},\Phi^*\mathbf{s}^{rec})$	$\phi^{rec}(\mathbf{x})=(\hat{\mathbf{x}},\Phi^*\mathbf{s}^{rec})$
penev_A_ E00116 (4.57)	022	■ 1.000	$\left\langle\mathbf{q}_m,\mathbf{s}_{\perp}\right\rangle=\left\langle\mathbf{q}_m,\mathbf{s}-\mathbf{s}_{\parallel}\right\rangle=\left\langle\mathbf{q}_m,\mathbf{s}\right\rangle-\left\langle\mathbf{q}_m,\mathbf{s}_{\parallel}\right\rangle=O_m-O_m=0.$	$\left\langle\mathbf{q}_m,\mathbf{s}_{\perp}\right\rangle=\left\langle\mathbf{q}_m,\mathbf{s}-\mathbf{s}_{\parallel}\right\rangle=\left\langle\mathbf{q}_m,\mathbf{s}\right\rangle-\left\langle\mathbf{q}_m,\mathbf{s}_{\parallel}\right\rangle=O_m-O_m=0.$	$\left\langle\mathbf{q}_m,\mathbf{s}_{\perp}\right\rangle=\left\langle\mathbf{q}_m,\mathbf{s}-\mathbf{s}_{\parallel}\right\rangle=\left\langle\mathbf{q}_m,\mathbf{s}\right\rangle-\left\langle\mathbf{q}_m,\mathbf{s}_{\parallel}\right\rangle=O_m-O_m=0.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00001	010	—	$T=15$	$T = 15$
penev_A_F00002	010	—	$T=87$	$T = 87$
penev_A_F00003	012	—	$T=1038$	$T = 1038$
penev_A_F00004	009	—	$(T=15)$	$(T = 15)$
penev_A_F00005		—	$(T=87)$	$(T = 87)$
penev_A_F00006	011	—	$S-U$	$S - U$
penev_A_F00007	012	—	$T=1038, \ V=3840$	$T = 1038, V = 3840$
penev_A_F00008	013	—	$1(T=1038, \ V=3840)$	$1(T = 1038, V = 3840)$
penev_A_F00009		—	$3.11 \ S-U$	$3.11S - U$
penev_A_F00010		—	$1(T=1038$	$1(T = 1038$
penev_A_F00011	007	—	$V=3840$	$V = 3840$
penev_A_F00012		—	$3.13 \ \log$	$3.13\log$
penev_A_F00013		—	$3.14 \ \mathrm{log}-\log$	$3.14\log - \log$
penev_A_F00014	014	—	$5(T=3279, \ V=3840)$	$5(T = 3279, V = 3840)$
penev_A_F00015	008	—	r	r
penev_A_F00016	014	—	$5(T=3279, \ V=3840), \ \log \ r$	$5(T = 3279, V = 3840), \log r$
penev_A_F00017		—	$T=1038, \ V=11520$	$T = 1038, V = 11520$
penev_A_F00018	016	—	$4(T=1038, \ V=11520)$	$4(T = 1038, V = 11520)$
penev_A_F00019		—	P	P
penev_A_F00020	018	—	$\mathrm{\mathbf{K}}$	K
penev_A_F00021	019	—	$\mathrm{\mathbf{P}}$	P
penev_A_F00022		—	$S-N$	$S - N$
penev_A_F00023		—	$\left(T_{\mathrm{\text{restricted}}}=3049\right)$	$(T_{\mathrm{restricted}} = 3049)$
penev_A_F00024	004	—	$(1959,1961 \ \mathrm{a})$	$(1959, 1961\mathrm{a})$
penev_A_F00025		—	H	H
penev_A_F00026	006	—	$\phi(x)$	$\phi(x)$
penev_A_F00027	005	—	$\sim 6 \times 10^6$	$\sim 6 \times 10^6$
penev_A_F00028	005	—	$\phi(\mathrm{\mathbf{x}})$	$\phi(\mathbf{x})$
penev_A_F00029	005	—	$\{\mathrm{\mathbf{x}}\}$	$\{\mathbf{x}\}$
penev_A_F00030	005	—	V	V
penev_A_F00031	005	—	$\{x\}$	$\{x\}$
penev_A_F00032		—	$V= \{x\} $	$V = \{x\} $
penev_A_F00033	005	—	$\mathrm{\mathbf{x}}$	x
penev_A_F00034	005	—	$\phi(\mathrm{\mathbf{x}})=I(\mathrm{\mathbf{x}})$	$\phi(\mathbf{x}) = I(\mathbf{x})$
penev_A_F00035	005	—	I	I
penev_A_F00036	005	—	$t,\{ \ }^4$	$t,^4$
penev_A_F00037	005	—	$p(t)$	$p(t)$
penev_A_F00038	005	—	$r(\theta, \ell)$	$r(\theta, \ell)$
penev_A_F00039	005	—	$\left\{\phi^i(\mathrm{\mathbf{x}}), \ t=1, \ \cdots, \ T\right\}$	$\left\{\phi^i(\mathbf{x}), t = 1, \cdots, T\right\}$
penev_A_F00040	005	—	T	T
penev_A_F00041	005	—	$\left\{\phi^t(\mathrm{\mathbf{x}})\right\}$	$\left\{\phi^t(\mathbf{x})\right\}$
penev_A_F00042	005	—	$\mathcal{P}[\phi(\mathrm{\mathbf{x}})]$	$\mathcal{P}[\phi(\mathbf{x})]$
penev_A_F00043	005	—	$\mathcal{K}$	$\mathcal{K}$
penev_A_F00044	005	—	$\left\{\mathcal{O}^t(\chi) \equiv \mathcal{K}\left[\phi^t(\mathrm{\mathbf{x}})\right]\right\}$	$\left\{\mathcal{O}^t(\chi) \equiv \mathcal{K}\left[\phi^t(\mathbf{x})\right]\right\}$
penev_A_F00045	005	—	$\{\chi\}$	$\{\chi\}$
penev_A_F00046	005	—	$\mathcal{O}$	$\mathcal{O}$
penev_A_F00047	005	—	$\{\mathrm{\mathbf{x}}\}$	$\{\mathbf{x}\}$
penev_A_F00048	005	—	$\mathcal{P}\left[\mathcal{O}^t(\chi)\right]$	$\mathcal{P}\left[\mathcal{O}^t(\chi)\right]$
penev_A_F00049		—	$ \{x\} $	$ \{x\} $
penev_A_F00050		—	$\phi^{\mathrm{\text{random}}}(\mathrm{\mathbf{x}})$	$\phi^{\mathrm{random}}(\mathbf{x})$
penev_A_F00051		—	$\mathcal{P}\left[\phi^{\mathrm{\text{random}}}(x)\right]$	$\mathcal{P}\left[\phi^{\mathrm{random}}(x)\right]$
penev_A_F00052		—	$V=60 \times 64 \approx 4000$	$V = 60 \times 64 \approx 4000$
penev_A_F00053	006	—	$\left\{\phi^t(\mathrm{\mathbf{x}})\right\}_1^T$	$\left\{\phi^t(\mathbf{x})\right\}_1^T$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00054	006	—	$R(\mathbf{x}, \mathbf{y})$	$R(\mathbf{x}, \mathbf{y})$
penev_A_F00055	006	—	$(\phi: 4000 \mapsto 100)$	$(\phi: 4000 \mapsto 100)$
penev_A_F00056	006	—	100^{4000}	100^{4000}
penev_A_F00057	006	—	$\phi(\mathbf{x})\left(\phi \in \mathbf{100}^{4000}\right)$	$\phi(\mathbf{x})\left(\phi \in \mathbf{100}^{4000}\right)$
penev_A_F00058	006	—	$\mathcal{P}[\phi(x)]$	$\mathcal{P}[\phi(x)]$
penev_A_F00059	006	—	$\mathcal{P}: 100^{4000} \mapsto 2\left(\mathcal{P} \in 2^{1000^{4000}}\right)$	$\mathcal{P}: 100^{4000} \mapsto 2\left(\mathcal{P} \in 2^{1000^{4000}}\right)$
penev_A_F00060	006	—	$100^{4000}=10^{8000}$	$100^{4000}=10^{8000}$
penev_A_F00061	006	—	$\sim 10^{7990}$	$\sim 10^{7990}$
penev_A_F00062	006	—	$\sim$	$\sim$
penev_A_F00063	006	—	5×10^{12}	5×10^{12}
penev_A_F00064	006	—	10^{11}	10^{11}
penev_A_F00065	006	—	10^{7978}	10^{7978}
penev_A_F00066	006	—	$\{ \}^8 \mathbf{R}$	$^8 \mathbf{R}$
penev_A_F00067	006	—	$(\mathbf{x}, \mathbf{y}) \in V^2$	$(\mathbf{x}, \mathbf{y}) \in V^2$
penev_A_F00068	006	—	V^2	V^2
penev_A_F00069	006	—	$R(\mathbf{x}, \mathbf{y})=R(\mathbf{y}, \mathbf{x})$	$R(\mathbf{x}, \mathbf{y})=R(\mathbf{y}, \mathbf{x})$
penev_A_F00070	006	—	$V=60 \times 64=3840$	$V=60 \times 64=3840$
penev_A_F00071	006	—	$7374,720 \approx 10^7$	$7374,720 \approx 10^7$
penev_A_F00072	006	—	$\phi^t(\mathbf{x})$	$\phi^t(\mathbf{x})$
penev_A_F00073	006	—	$\phi^t: V \mapsto \mathcal{R}$	$\phi^t: V \mapsto \mathcal{R}$
penev_A_F00074	006	—	t	t
penev_A_F00075	006	—	$\phi^i \in \mathcal{R}^V \equiv U$	$\phi^i \in \mathcal{R}^V \equiv U$
penev_A_F00076	006	—	$\phi \in U$	$\phi \in U$
penev_A_F00077	006	—	$\left\{\phi^t(x), t=1, \cdots, T\right\}$	$\left\{\phi^t(x), t=1, \cdots, T\right\}$
penev_A_F00078	006	—	$t=1, \cdots, T$	$t=1, \cdots, T$
penev_A_F00079	006	—	$S \equiv \mathcal{R}^T$	$S \equiv \mathcal{R}^T$
penev_A_F00080	006	—	$\hat{\mathbf{x}} \in U$	$\hat{\mathbf{x}} \in U$
penev_A_F00081	006	—	$\hat{t} \in S$	$\hat{t} \in S$
penev_A_F00082	006	—	U	U
penev_A_F00083	006	—	S	S
penev_A_F00084	006	—	$\phi \equiv \mathbf{1}_y \phi=\frac{1}{V} \sum_x \hat{\mathbf{x}}(\hat{\mathbf{x}}, \phi)$	$\phi \equiv \mathbf{1}_y \phi=\frac{1}{V} \sum_x \hat{\mathbf{x}}(\hat{\mathbf{x}}, \phi)$
penev_A_F00085	006	—	$\hat{\mathbf{x}}, \phi$	$\hat{\mathbf{x}}, \phi$
penev_A_F00086	006	—	$\mathcal{R}$	$\mathcal{R}$
penev_A_F00087	006	—	$\mathcal{R}^V$	$\mathcal{R}^V$
penev_A_F00088	006	—	ϕ	ϕ
penev_A_F00089		—	(2.9)	(2.9)
penev_A_F00090	006	—	$\ \phi\ _U^2$	$\ \phi\ _U^2$
penev_A_F00091	006	—	ϕ'^{11}	ϕ'^{11}
penev_A_F00092	006	—	$\langle \mathbf{s}, \Phi \mathbf{u} \rangle$	$\langle \mathbf{s}, \Phi \mathbf{u} \rangle$
penev_A_F00093	006	—	Φ	Φ
penev_A_F00094	006	—	$\mathbf{s}=\mathbf{s}_0$	$\mathbf{s}=\mathbf{s}_0$
penev_A_F00095	006	—	$\mathbf{u} \in U$	$\mathbf{u} \in U$
penev_A_F00096	006	—	$\langle \mathbf{s}_0, \Phi \mathbf{u} \rangle$	$\langle \mathbf{s}_0, \Phi \mathbf{u} \rangle$
penev_A_F00097	006	—	$\mathbf{v}=\Phi^*(\mathbf{s}) \in U$	$\mathbf{v}=\Phi^*(\mathbf{s}) \in U$
penev_A_F00098	006	—	$\langle \mathbf{s}, \Phi \mathbf{u} \rangle=(\mathbf{v}, \mathbf{u})$	$\langle \mathbf{s}, \Phi \mathbf{u} \rangle=(\mathbf{v}, \mathbf{u})$
penev_A_F00099	006	—	$\Phi^*(\mathbf{s})$	$\Phi^*(\mathbf{s})$
penev_A_F00100	006	—	$\Phi^*: S \mapsto U$	$\Phi^*: S \mapsto U$
penev_A_F00101	006	—	Φ^*	Φ^*
penev_A_F00102	006	—	$\Phi: U \mapsto S$	$\Phi: U \mapsto S$
penev_A_F00103	006	—	$\Phi(2.12), \Phi \hat{\mathbf{x}} \equiv \mathbf{1}_s \Phi \hat{\mathbf{x}}=\frac{1}{T} \sum_t \hat{t}(\hat{t}, \Phi \hat{\mathbf{x}})$	$\Phi(2.12), \Phi \hat{\mathbf{x}} \equiv \mathbf{1}_s \Phi \hat{\mathbf{x}}=\frac{1}{T} \sum_t \hat{t}(\hat{t}, \Phi \hat{\mathbf{x}})$
penev_A_F00104	006	—	$\cos ^2(\phi, \phi')=\frac{\left \left(\phi, \phi'\right)\right ^2}{(\phi, \phi)\left(\phi', \phi'\right)}=\frac{\left \left(\phi, \phi'\right)\right ^2}{\ \phi\ '\left\ \phi'\right\ v} \in[0,1]$	$\cos ^2(\phi, \phi')=\frac{\left \left(\phi, \phi'\right)\right ^2}{(\phi, \phi)\left(\phi', \phi'\right)}=\frac{\left \left(\phi, \phi'\right)\right ^2}{\ \phi\ '\left\ \phi'\right\ v} \in[0,1]$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00105	006	—	$\left(\Phi^*\right)^*=\Phi$	$(\Phi^*)^* = \Phi$
penev_A_F00106	006	—	$\left\langle \mathrm{left\langl\mathbf{s},\mathrm{left(\Phi^*\right)^*\mathbf{u}\right\rangle =\left(\Phi^*\mathbf{s},\mathbf{u}\right)$	$\langle \mathbf{s}, (\Phi^*)^* \mathbf{u} \rangle = (\Phi^* \mathbf{s}, \mathbf{u})$
penev_A_F00107	006	—	$\mathrm{left\langl\mathbf{s},\Phi\mathbf{u}\right\rangle =\left(\Phi^*\mathbf{u}\right)$	$\langle \mathbf{s}, \Phi \mathbf{u} \rangle = \langle \mathbf{s}, (\Phi^*)^* \mathbf{u} \rangle$
penev_A_F00108	006	—	$\mathbf{s}$	$\mathbf{s}$
penev_A_F00109	006	—	$\mathbf{u}$	$\mathbf{u}$
penev_A_F00110	006	—	$\hat{\mathbf{x}}$	$\hat{\mathbf{x}}$
penev_A_F00111	006	—	$\{t\}$	$\{t\}$
penev_A_F00112	006	—	$s\in S$	$s \in S$
penev_A_F00113	006	—	$\mathbf{y}$	$\mathbf{y}$
penev_A_F00114	006	—	$\Phi\hat{\mathbf{x}}\in S$	$\Phi \hat{\mathbf{x}} \in S$
penev_A_F00115	006	—	$\Phi\hat{\mathbf{y}}\in S$	$\Phi \hat{\mathbf{y}} \in S$
penev_A_F00116	006	—	$\Phi^*\hat{t}$	$\Phi^* \hat{t}$
penev_A_F00117	006	—	$s\in S$	$s \in S$
penev_A_F00118	006	—	$\Phi^*\mathbf{s}\in U$	$\Phi^* \mathbf{s} \in U$
penev_A_F00119	006	—	$\{\phi^t\equiv\Phi^*\hat{t}\}_{t=1}^T$	$\{\phi^t \equiv \Phi^* \hat{t}\}_{t=1}^T$
penev_A_F00120	006	—	$\{\mathcal{P}\left[\Phi^*t\right]\}$	$\{\mathcal{P}\left[\Phi^*t\right]\}$
penev_A_F00121	006	—	${}^{13}(\Phi\hat{\mathbf{x}},\Phi\hat{\mathbf{y}})=(\hat{\mathbf{x}},\Phi^*\Phi\hat{\mathbf{y}})=(\hat{\mathbf{x}},\mathbf{R}\hat{\mathbf{y}})=R(\mathbf{x},\mathbf{y})$	${}^{13}(\Phi \hat{\mathbf{x}}, \Phi \hat{\mathbf{y}}) = (\hat{\mathbf{x}}, \Phi^* \Phi \hat{\mathbf{y}}) = (\hat{\mathbf{x}}, \mathbf{R} \hat{\mathbf{y}}) = R(\mathbf{x}, \mathbf{y})$
penev_A_F00122	006	—	$(x,\,y)$	(x, y)
penev_A_F00123	006	—	$\mathbf{R}\equiv\Phi^*\Phi(2.1)$	$\mathbf{R} \equiv \Phi^* \Phi (2.1)$
penev_A_F00124	006	—	$\mathrm{tr}\,\mathbf{R}\equiv\sum_{\mathbf{x}}\frac{(\hat{x},\mathbf{h}\hat{x})}{(\hat{x},\hat{x})}=\frac{1}{V}\sum_{\mathbf{x}}(\hat{x},\mathbf{R}\hat{\mathbf{x}})=$	$\mathrm{tr} \mathbf{R} \equiv \sum_{\mathbf{x}} \frac{(\hat{x}, \mathbf{h} \hat{x})}{(\hat{x}, \hat{x})} = \frac{1}{V} \sum_{\mathbf{x}} (\hat{x}, \mathbf{R} \hat{\mathbf{x}}) =$
penev_A_F00125	006	—	$\frac{1}{V}\sum_{\mathbf{x}}\frac{1}{T}\sum_t \phi^t(\mathbf{x}) ^2$	$\frac{1}{V} \sum_{\mathbf{x}} \frac{1}{T} \sum_t \phi^t(\mathbf{x}) ^2$
penev_A_F00126	006	—	$\mathbf{R}$	$\mathbf{R}$
penev_A_F00127	006	—	$\mathrm{tr}\,\mathbf{R}=\mathrm{tr}\,\Phi^*\Phi$	$\mathrm{tr} \mathbf{R} = \mathrm{tr} \Phi^* \Phi$
penev_A_F00128	007	—	$\Phi^*s=\phi$	$\Phi^* s = \phi$
penev_A_F00129	007	—	$\mathbf{s}(\phi)\in S$	$\mathbf{s}(\phi) \in S$
penev_A_F00130	007	—	$\mathcal{P}[\phi]$	$\mathcal{P}[\phi]$
penev_A_F00131	007	—	$\phi=\Phi^*s$	$\phi = \Phi^* s$
penev_A_F00132	007	—	$\Phi^{*+}\Phi^*\mathbf{s}=\mathbf{s}$	$\Phi^{*+} \Phi^* \mathbf{s} = \mathbf{s}$
penev_A_F00133	007	—	$s\in S.^{17}$	$s \in S.^{17}$
penev_A_F00134	007	—	Φ^{*+}	Φ^{*+}
penev_A_F00135	007	—	$\mathbf{s}=\Phi^*\phi$	$\mathbf{s} = \Phi^* \phi$
penev_A_F00136	007	—	$\Phi^*\mathbf{s}=$	$\Phi^* \mathbf{s} =$
penev_A_F00137	007	—	$\Phi^*\Phi^{*+}\phi=\phi$	$\Phi^* \Phi^{*+} \phi = \phi$
penev_A_F00138	007	—	Φ,Φ^*	Φ, Φ^*
penev_A_F00139	007	—	$T\leq V$	$T \leq V$
penev_A_F00140	007	—	${}^{16}\Phi^{*+}$	${}^{16} \Phi^{*+}$
penev_A_F00141	007	—	$T>V$	$T > V$
penev_A_F00142	007	—	$T\in[16,3098]$	$T \in [16, 3098]$
penev_A_F00143	007	—	$T>20,000,000$	$T > 20,000,000$
penev_A_F00144	007	—	Φ^*t	$\Phi^* t$
penev_A_F00145	007	—	$\Phi^{*+}\Phi^*\hat{t}\neq\hat{t}$	$\Phi^{*+} \Phi^* \hat{t} \neq \hat{t}$
penev_A_F00146	007	—	$\mathcal{P}\left[\Phi^*s\right]\propto\exp\left\{-\frac{1}{2}\left\ \Phi^{*+}\Phi^*s\right\ _S^2\right\}$	$\mathcal{P}\left[\Phi^*s\right] \propto \exp\left\{-\frac{1}{2}\left\ \Phi^{*+}\Phi^*s\right\ _S^2\right\}$
penev_A_F00147	007	—	$\Phi^*\Phi$	$\Phi^* \Phi$
penev_A_F00148	007	—	$\Phi^*\Phi\mathbf{u}=0\Rightarrow\left\ \Phi\mathbf{u}\right\ _S^2=0\Rightarrow\Phi\mathbf{u}=0$	$\Phi^* \Phi \mathbf{u} = 0 \Rightarrow \left\ \Phi \mathbf{u} \right\ _S^2 = 0 \Rightarrow \Phi \mathbf{u} = 0$
penev_A_F00149	007	—	$\ker\Phi^*\Phi\subset\ker\Phi$	$\ker \Phi^* \Phi \subset \ker \Phi$
penev_A_F00150	007	—	$\Phi\mathbf{u}=0\Rightarrow\Phi^*\Phi\mathbf{u}=\Phi^*0=0$	$\Phi \mathbf{u} = 0 \Rightarrow \Phi^* \Phi \mathbf{u} = \Phi^* 0 = 0$
penev_A_F00151	007	—	$\ker\Phi\subset\ker\Phi^*\Phi$	$\ker \Phi \subset \ker \Phi^* \Phi$
penev_A_F00152	007	—	$\{\psi_\sigma\}$	$\{\psi_\sigma\}$
penev_A_F00153	007	—	$\{\sigma^2\geq 0\}$	$\{\sigma^2 \geq 0\}$
penev_A_F00154	007	—	$\{\psi_a\mid\sigma=0\}$	$\{\psi_a \mid \sigma = 0\}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00155	007	—	$\operatorname{ker} \Phi^* \Phi$	$\ker \Phi^* \Phi$
penev_A_F00156	007	—	$\operatorname{ker} \Phi$	$\ker \Phi$
penev_A_F00157	007	—	$\left\{ \psi_\sigma \mid \sigma > 0 \right\}$	$\{\psi_\sigma \mid \sigma > 0\}$
penev_A_F00158	007	—	$\left(\Phi^* \right)$	(Φ^*)
penev_A_F00159	007	—	$\mathbf{P} = \mathbf{P}_{\left(\left(\operatorname{ker} \Phi^* \right)^\perp \right)} = \mathbf{P}_{\left(\operatorname{ker} \Phi \right)^\perp}$	$\mathbf{P} = \mathbf{P}_{\operatorname{range}(\Phi^*)} = \mathbf{P}_{(\ker \Phi)^\perp}$
penev_A_F00160	007	—	$\Phi \left(\operatorname{ker} \Phi \right)^\perp \in U$	$\Phi(\ker \Phi)^\perp \in U$
penev_A_F00161	007	—	$\operatorname{range}(\Phi) \in S$	$\operatorname{range}(\Phi) \in S$
penev_A_F00162	007	—	$\sigma > 0$	$\sigma > 0$
penev_A_F00163	007	—	$\operatorname{range}(\Phi) =$	$\operatorname{range}(\Phi) =$
penev_A_F00164	007	—	$\left\{ \mathbf{s}_\sigma \right\}$	$\{\mathbf{s}_\sigma\}$
penev_A_F00165	007	—	$\sigma \sigma' \left(\mathbf{s}_\sigma, \mathbf{s}_{\sigma'} \right) = \left\langle \Phi \psi_\sigma, \Phi \psi_{\sigma'} \right\rangle = \left(\psi_\sigma, \Phi \Phi^* \psi_{\sigma'} \right) =$	$\sigma \sigma' \left(\mathbf{s}_\sigma, \mathbf{s}_{\sigma'} \right) = \left\langle \Phi \psi_\sigma, \Phi \psi_{\sigma'} \right\rangle = \left(\psi_\sigma, \Phi \Phi^* \psi_{\sigma'} \right) =$
penev_A_F00166	007	—	$\sigma^2 \delta_{\sigma_x \sigma'}$	$\sigma^2 \delta_{\sigma_x \sigma'}$
penev_A_F00167	007	—	$\left\{ s_\sigma \right\}$	$\{s_\sigma\}$
penev_A_F00168	007	—	$\operatorname{range}(\Phi)$	$\operatorname{range}(\Phi)$
penev_A_F00169	007	—	$\Phi^{*+} \Phi^* \left(S \xrightarrow{\Phi^*} U \xrightarrow{\Phi^{*+}} S \right)$	$\Phi^{*+} \Phi^* \left(S \xrightarrow{\Phi^*} U \xrightarrow{\Phi^{*+}} S \right)$
penev_A_F00170	007	—	$\operatorname{range} \left(\Phi^{*+} \right)$	$\operatorname{range} \left(\Phi^{*+} \right)$
penev_A_F00171	007	—	$\mathbf{P}' : S \mapsto S$	$\mathbf{P}' : S \mapsto S$
penev_A_F00172	007	—	$\mathbf{P} : U \mapsto U(2.26)$	$\mathbf{P} : U \mapsto U(2.26)$
penev_A_F00173	007	—	$\Phi^* \Phi^{*+} \left(U^{\Phi^{**}} S \xrightarrow{\Phi^*} U \right)$	$\Phi^* \Phi^{*+} \left(U^{\Phi^{**}} S \xrightarrow{\Phi^*} U \right)$
penev_A_F00174	007	—	$\mathbf{P}'(2.33)$	$\mathbf{P}'(2.33)$
penev_A_F00175	007	—	s	s
penev_A_F00176	007	—	$\mathbf{P}'$	$\mathbf{P}'$
penev_A_F00177		—	$\left(\mathbf{P}' \hat{t} = \hat{t} \right)$	$(\mathbf{P}' \hat{t} = \hat{t})$
penev_A_F00178		—	$\mathbf{P}' \hat{t} \in S$	$\mathbf{P}' \hat{t} \in S$
penev_A_F00179	008	—	$\{ \hat{t} \}$	$\{\hat{t}\}$
penev_A_F00180	008	—	$\Phi^* \hat{t} \in U$	$\Phi^* \hat{t} \in U$
penev_A_F00181	008	—	$\mathbf{P}' \hat{t} \in S^{22}$	$\mathbf{P}' \hat{t} \in S^{22}$
penev_A_F00182	008	—	a_σ^t	a_σ^t
penev_A_F00183	008	—	$\hat{t}, \phi^t = \Phi^* \hat{t} = \Phi^* \mathbf{P}' \hat{t} = \sum_{\sigma > 0} a_\sigma^t \Phi^* \mathbf{s}_\sigma = \sum_{\sigma > 0} \sigma a_\sigma^t \psi_\sigma$	$\hat{t}, \phi^t = \Phi^* \hat{t} = \Phi^* \mathbf{P}' \hat{t} = \sum_{\sigma > 0} a_\sigma^t \Phi^* \mathbf{s}_\sigma = \sum_{\sigma > 0} \sigma a_\sigma^t \psi_\sigma$
penev_A_F00184	008	—	ϕ^t	ϕ^t
penev_A_F00185		—	(2.40)	(2.40)
penev_A_F00186	008	—	$1s(2.5)$	$1s(2.5)$
penev_A_F00187	008	—	ψ_σ	ψ_σ
penev_A_F00188	008	—	$\Phi \psi_\sigma = \sigma \mathbf{s}_\sigma = \sigma \mathbf{1}_S \mathbf{s}_\sigma = \frac{1}{T} \sum_t \hat{t} \left\langle \hat{t}, \sigma \mathbf{s}_\sigma \right\rangle =$	$\Phi \psi_\sigma = \sigma \mathbf{s}_\sigma = \sigma \mathbf{1}_S \mathbf{s}_\sigma = \frac{1}{T} \sum_t \hat{t} \left\langle \hat{t}, \sigma \mathbf{s}_\sigma \right\rangle =$
penev_A_F00189	008	—	$\frac{1}{T} \sum_t \sigma a_\sigma^t \hat{t}$	$\frac{1}{T} \sum_t \sigma a_\sigma^t \hat{t}$
penev_A_F00190	008	—	$\{ \sigma \mathbf{s}_\sigma \}$	$\{\sigma \mathbf{s}_\sigma\}$
penev_A_F00191	008	—	$\left(\psi_\sigma, \mathbf{R} \psi_{\sigma'} \right) = \sigma^2 \delta_{\sigma, \sigma'}$	$(\psi_\sigma, \mathbf{R} \psi_{\sigma'}) = \sigma^2 \delta_{\sigma, \sigma'}$
penev_A_F00192	008	—	$a_\sigma^t \equiv \left(\mathbf{s}_\sigma, \hat{t} \right)$	$a_\sigma^t \equiv (\mathbf{s}_\sigma, \hat{t})$
penev_A_F00193	008	—	$\mathbf{s}_\sigma = \mathbf{1}_S \mathbf{s}_\sigma = \frac{1}{T} \sum_t \hat{t} \left\{ \hat{t}, \mathbf{s}_\sigma \right\} = \frac{1}{T} \sum_t a_\sigma^t \hat{t}$	$\mathbf{s}_\sigma = \mathbf{1}_S \mathbf{s}_\sigma = \frac{1}{T} \sum_t \hat{t} \left\{ \hat{t}, \mathbf{s}_\sigma \right\} = \frac{1}{T} \sum_t a_\sigma^t \hat{t}$
penev_A_F00194	008	—	$\psi_\sigma = \sigma^{-1} \Phi'' \mathbf{s}_\sigma =$	$\psi_\sigma = \sigma^{-1} \Phi'' \mathbf{s}_\sigma =$
penev_A_F00195	008	—	$\frac{1}{T} \sum_t a_\sigma^i \Phi^{* \hat{i}} = \frac{1}{T} \sum_t a_\sigma^t \phi^t$	$\frac{1}{T} \sum_t a_\sigma^i \Phi^{* \hat{i}} = \frac{1}{T} \sum_t a_\sigma^t \phi^t$
penev_A_F00196	008	—	$T < V$	$T < V$
penev_A_F00197	008	—	$\Phi \dot{\mathbf{x}}$	$\Phi \dot{\mathbf{x}}$
penev_A_F00198	008	—	$\Phi \psi_\sigma$	$\Phi \psi_\sigma$
penev_A_F00199	008	—	$\{ \sigma \}$	$\{\sigma\}$
penev_A_F00200	008	—	$\{ \sigma_r \},^{23}$	$\{\sigma_r\},^{23}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00201	008	—	$r>r^{\mathrm{\prime}}\rightarrow\sigma_r\leq\sigma_{r^{\mathrm{\prime}}}$	$r>r^{\prime}\Rightarrow\sigma_r\leq\sigma_{r^{\prime}}$
penev_A_F00202	008	—	$\psi_r\equiv\psi_{\sigma_r}$	$\psi_r\equiv\psi_{\sigma_r}$
penev_A_F00203	008	—	$z_r\equiv\mathrm{\bm{s}}_{\sigma_\rho}$	$z_r\equiv\mathbf{s}_{\sigma_\rho}$
penev_A_F00204	008	—	$\mathrm{\bm{R}}\equiv\Phi^*\Phi_1$	$\mathbf{R}\equiv\Phi^*\Phi_1$
penev_A_F00205	008	—	$\operatorname{tr}R$	$\mathrm{tr}R$
penev_A_F00206	008	—	N	N
penev_A_F00207	008	—	Φ_N	Φ_N
penev_A_F00208	008	—	$\frac{\operatorname{tr}\mathrm{\bm{R}}_N}{\operatorname{tr}\mathrm{\bm{R}}}\in[0,1]$	$\frac{\mathrm{tr}\mathbf{R}_N}{\mathrm{tr}\mathbf{R}}\in[0,1]$
penev_A_F00209	008	—	T/N	T/N
penev_A_F00210	008	—	$-\log_2\frac{\left \left \phi^{\mathrm{err}}\right \right ^2}{\left \phi\right ^2}.^{24}$	$-\log_2\frac{\ \phi^{\mathrm{err}}\ ^2}{\ \phi\ ^2}.^{24}$
penev_A_F00211	008	—	$\log_{10}2\approx0.301$	$\log_{10}2\approx0.301$
penev_A_F00212	008	—	$N<T\ll V$	$N<T\ll V$
penev_A_F00213	009	—	$T=16$	$T=16$
penev_A_F00214	009	—	$\left\{\phi^t(\mathbf{x})\right\}_{t=1}^{15}$	$\{\phi^t(\mathbf{x})\}_{t=1}^{15}$
penev_A_F00215	009	—	$\left\{\Psi_r(\mathbf{x})\equiv\left(\hat{\mathbf{x}},\psi_r\right)\right\}_{r=1}^{15}$	$\{\Psi_r(\mathbf{x})\equiv(\hat{\mathbf{x}},\psi_r)\}_{r=1}^{15}$
penev_A_F00216	009	—	Ψ_1	Ψ_1
penev_A_F00217	009	—	$\left(\psi_r=\frac{1}{T}\sum_t a_r^t\phi^t\right)$	$\left(\psi_r=\frac{1}{T}\sum_t a_r^t\phi^t\right)$
penev_A_F00218	009	—	ϕ^{11}	ϕ^{11}
penev_A_F00219	009	—	Ψ_4	Ψ_4
penev_A_F00220	009	—	Ψ_2	Ψ_2
penev_A_F00221	009	—	a_r^t	a_r^t
penev_A_F00222	009	—	$\Psi_r(\mathbf{x})$	$\Psi_r(\mathbf{x})$
penev_A_F00223	009	—	(Ψ_1)	(Ψ_1)
penev_A_F00224	009	—	$(\Psi_2)^.^3$	$(\Psi_2)^.^3$
penev_A_F00225	009	—	(Ψ_3)	(Ψ_3)
penev_A_F00226	009	—	ϕ_N^{rec}	ϕ_N^{rec}
penev_A_F00227	009	—	$\left\{\phi_N^{\mathrm{rec}}\right\}_{N=1}^T$	$\{\phi_N^{\mathrm{rec}}\}_{N=1}^T$
penev_A_F00228	009	—	ϕ^{12}	ϕ^{12}
penev_A_F00229	009	—	$-\log_2\frac{\left \left \phi^{ev}\mathbf{r}\right \right ^2}{\left \phi\right ^2}$	$-\log_2\frac{\ \phi^{ev}\mathbf{r}\ ^2}{\ \phi\ ^2}$
penev_A_F00230	009	—	$\phi_N^{\mathrm{err}}\equiv\phi-\phi_N^{\mathrm{rec}}=\left(\mathbf{1}-\mathbf{P}_N\right)\phi$	$\phi_N^{\mathrm{err}}\equiv\phi-\phi_N^{\mathrm{rec}}=(\mathbf{1}-\mathbf{P}_N)\phi$
penev_A_F00231	009	—	$N>12$	$N>12$
penev_A_F00232	009	—	(ϕ^{rec})	(ϕ^{rec})
penev_A_F00233	009	—	ϕ^{err}	ϕ^{err}
penev_A_F00234	009	—	≈6	≈6
penev_A_F00235	009	—	$2^{-6}\approx1.5\%$	$2^{-6}\approx1.5\%$
penev_A_F00236	010	—	$\mathrm{SNR}=6.14$	$\mathrm{SNR}=6.14$
penev_A_F00237	010	—	$\mathrm{SNR}\approx8$	$\mathrm{SNR}\approx8$
penev_A_F00238	010	—	Ψ_{13}	Ψ_{13}
penev_A_F00239	010	—	$\left\{\sigma_r\right\}_{r=1}^N(2.52)$	$\{\sigma_r\}_{r=1}^N(2.52)$
penev_A_F00240	010	—	$-\log\mathcal{P}[\phi]$	$-\log\mathcal{P}[\phi]$
penev_A_F00241	010	—	~3.8	~3.8
penev_A_F00242	010	—	≈4	≈4
penev_A_F00243	010	—	$\left\{\Psi_r(\mathbf{x})\right\}_{r=1}^{16}$	$\{\Psi_r(\mathbf{x})\}_{r=1}^{16}$
penev_A_F00244	010	—	$V=96\times120=11,520$	$V=96\times120=11,520$
penev_A_F00245	010	—	Ψ_3	Ψ_3
penev_A_F00246	010	—	ϕ^{rec}	ϕ^{rec}
penev_A_F00247	010	—	T/V	T/V
penev_A_F00248	010	—	≈9.5	≈9.5
penev_A_F00249	010	—	$N=100$	$N=100$
penev_A_F00250	010	—	$T\approx200$	$T\approx200$
penev_A_F00251	010	—	$V=96\times90=8,640$	$V=96\times90=8,640$
Conf. MathPix confidence: ■ ≥0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00252	010	—	V=64 \times 60=3,840	$V = 64 \times 60 = 3,840$
penev_A_F00253	010	—	3.3 B	$3.3B$
penev_A_F00254	010	—	\phi^{e\,r\,r}	ϕ^{err}
penev_A_F00255	010	—	\mathrm{SNR} \approx 5.7	$\mathrm{SNR} \approx 5.7$
penev_A_F00256	011	—	\Psi_{1}, \Psi_{3}	Ψ_1, Ψ_3
penev_A_F00257	011	—	\boldsymbol{\Psi}_{\mathbf{1}}	$\boldsymbol{\Psi}_1$
penev_A_F00258	011	—	\boldsymbol{\Psi}_{\mathbf{1}}	$\boldsymbol{\Psi}_1$
penev_A_F00259	011	—	\log _{2}(1-	$\log _2(1-$
penev_A_F00260	011	—	)=\log _{2}\left(\frac{\text { trfl-trfu }}{\text { trf }}\right)	$)=\log _2\left(\frac{\text {trfl-trfu}}{\text {trf}}\right)$
penev_A_F00261	011	—	\langle S\rangle=\frac{1}{N}\left \left\langle\left(\Phi_N\right)^{\dagger} \phi\right\rangle_S\right ^2	$\langle S\rangle=\frac{1}{N}\left\ \left(\Phi_N\right)^{* \dagger} \phi\right\ _S^2$
penev_A_F00262	011	—	\langle S\rangle	$\langle S\rangle$
penev_A_F00263	011	—	\mathcal{P}[\phi] \propto \exp \left\{-\frac{1}{2}\langle S\rangle\right\},\langle S\rangle	$\mathcal{P}[\phi] \propto \exp \left\{-\frac{1}{2}\langle S\rangle\right\},\langle S\rangle$
penev_A_F00264	011	—	{ }^9,\langle S\rangle	$^9, \langle S\rangle$
penev_A_F00265	011	—	\approx 0.3	≈ 0.3
penev_A_F00266	011	—	1(T=1038)	$1(T=1038)$
penev_A_F00267	011	—	(V=3840)	$(V=3840)$
penev_A_F00268	011	—	a_{2}	a_2
penev_A_F00269		—	\pi	π
penev_A_F00270		—	\mathbf{\Psi}_{14}	$\square_{14}$
penev_A_F00271		—	\Psi_{14}	Ψ_{14}
penev_A_F00272	012	—	\left\{\Psi_r(\mathbf{x})\right\}_{\tau=1}^{16}	$\left\{\Psi_r(\mathbf{x})\right\}_{\tau=1}^{16}$
penev_A_F00273	012	—	\Psi_5	Ψ_5
penev_A_F00274	012	—	\left(\Psi_{12}, \Psi_{19}, \Psi_{22}\right)	$(\Psi_{12}, \Psi_{19}, \Psi_{22})$
penev_A_F00275	012	—	\Psi_{50}	Ψ_{50}
penev_A_F00276	012	—	N=300	$N=300$
penev_A_F00277	012	—	N=500	$N=500$
penev_A_F00278	012	—	N=1000 .{ }^{15}	$N=1000.^{15}$
penev_A_F00279	012	—	\mathrm{SNR} \approx 8.5	$\mathrm{SNR} \approx 8.5$
penev_A_F00280	012	—	N \sim 70	$N \sim 70$
penev_A_F00281	012	—	\mathrm{SNR} \approx 6	$\mathrm{SNR} \approx 6$
penev_A_F00282	012	—	\mathrm{SNR} \approx 7, N \sim 170	$\mathrm{SNR} \approx 7, N \sim 170$
penev_A_F00283	012	—	\mathrm{SNR} \approx 7.5, N \sim 350	$\mathrm{SNR} \approx 7.5, N \sim 350$
penev_A_F00284	012	—	N=	$N=$
penev_A_F00285	012	—	N=350	$N=350$
penev_A_F00286	012	—	T>3000	$T>3000$
penev_A_F00287	012	—	\Psi_{700}	Ψ_{700}
penev_A_F00288	012	—	\Psi_{800}	Ψ_{800}
penev_A_F00289	012	—	p	p
penev_A_F00290	012	—	L \times p	$L \times p$
penev_A_F00291	012	—	L	L
penev_A_F00292	012	—	p+1	$p+1$
penev_A_F00293	012	—	\Psi_{22} \Psi_{500}	$\Psi_{22} \Psi_{500}$
penev_A_F00294	012	—	\sin (k x)	$\sin (k x)$
penev_A_F00295	012	—	\cos (k x)	$\cos (k x)$
penev_A_F00296	012	—	\Psi_{1000}	Ψ_{1000}
penev_A_F00297	012	—	\frac{\left C v_r \sigma_r^2 v_r\right ^2}{\left v_r\right ^2}	$\frac{\left\ C v_r \sigma_r^2 v_r\right\ ^2}{\left v_r\right ^2}$
penev_A_F00298	012	—	\log _{2}	$\log _2$
penev_A_F00299	012	—	\sim -15	~ -15
penev_A_F00300	012	—	\log _{2} \frac{\sigma_r^2}{\sigma^2}	$\log _2 \frac{\sigma_r^2}{\sigma^2}$
penev_A_F00301	012	—	\sim -30	~ -30
penev_A_F00302	012	—	1,\log _{2}\left(1-\left \mathbf{v}_{\mathrm{r}}\right ^2\right)	$1, \log _2\left(1-\left\ \mathbf{v}_r\right\ ^2\right)$
penev_A_F00303	012	—	\mathrm{SNR} \approx 7.5	$\mathrm{SNR} \approx 7.5$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00304	012	—	$\phi_{\mathrm{N}}^{\mathrm{r a x}}$	$\phi_N^{r a x}$
penev_A_F00305	012	—	$\phi_{\mathrm{N}}^{\mathrm{e r r}}$	$\phi_N^{e r r}$
penev_A_F00306	012	—	$\mathrm{SNR} \approx 6.5$	$\mathrm{SNR} \approx 6.5$
penev_A_F00307	012	—	$\mathrm{SNR} \approx 7$	$\mathrm{SNR} \approx 7$
penev_A_F00308	012	—	$\mathrm{SNR} \approx 7.8$	$\mathrm{SNR} \approx 7.8$
penev_A_F00309	012	—	$N \in [300, 400]$	$N \in [300, 400]$
penev_A_F00310	013	—	$\phi_{\mathrm{N}}^{\mathrm{\text{err}}}$	ϕ_N^{err}
penev_A_F00311	013	—	$5 \times$	$5 \times$
penev_A_F00312	013	—	$-20 \times$	$-20 \times$
penev_A_F00313	013	—	$1 \, \mathrm{B}$	$1B$
penev_A_F00314	013	—	$(N=500)$	$(N = 500)$
penev_A_F00315	013	—	$(N=1000)$	$(N = 1000)$
penev_A_F00316	013	—	$-\alpha$	$-\alpha$
penev_A_F00317	013	—	$\{ \}^{16} \, \mathrm{By}$	$^{16}\mathrm{By}$
penev_A_F00318	016	—	$\log$	$\log$
penev_A_F00319	013	—	Ψ_{N}	Ψ_N
penev_A_F00320	013	—	$\log_{\{2\}} \frac{\sigma_{\mathrm{r}}^2}{\mathrm{tr} \mathbf{R}} \frac{\sigma_{\mathrm{r}}^2}{\mathrm{tr} \mathbf{R}}$	$\log_2 \frac{\sigma_r^2}{\mathrm{tr} \mathbf{R}} \frac{\sigma_r^2}{\mathrm{tr} \mathbf{R}}$
penev_A_F00321	013	—	$\log_{\{2\}} \frac{\operatorname{operatorname{tr}} \mathrm{R} - \operatorname{operatorname{tr}} \mathrm{R}}{\operatorname{operatorname{tr}} \mathrm{R}} = \log_2 (1 - \frac{\operatorname{operatorname{tr}} \mathrm{R} - \operatorname{operatorname{tr}} \mathrm{R}}{\operatorname{operatorname{tr}} \mathrm{R}})$	$\log_2 \frac{\mathrm{tr} \mathbf{R} - \mathrm{tr} \mathbf{R}_N}{\mathrm{tr} \mathbf{R}} = \log_2 (1 - \frac{\mathrm{tr} \mathbf{R} - \mathrm{tr} \mathbf{R}_N}{\mathrm{tr} \mathbf{R}})$
penev_A_F00322	013	—	$\Phi \Phi^*$	$\Phi \Phi^*$
penev_A_F00323	013	—	$\log - \log$	$\log - \log$
penev_A_F00324	013	—	$N=400$	$N = 400$
penev_A_F00325	013	—	$N(2.45)$	$N(2.45)$
penev_A_F00326	013	—	K	K
penev_A_F00327	013	—	$\left\{ \phi^k \right\}_{k \in K}$	$\{\phi^k\}_{k \in K}$
penev_A_F00328	013	—	$\left\{ \mathbf{P}_{\mathrm{N}} \phi^k \right\}_{k \in K}$	$\{\mathbf{P}_N \phi^k\}_{k \in K}$
penev_A_F00329	013	—	$\mathbf{P}_{\mathrm{N}} \phi$	$\mathbf{P}_N \phi$
penev_A_F00330	013	—	k	k
penev_A_F00331	013	—	$\mathbf{P}_{\mathrm{N}} \phi^k$	$\mathbf{P}_N \phi^k$
penev_A_F00332	013	—	$\mathbf{P}_{\mathrm{N}} \phi(2.8)$	$\mathbf{P}_N \phi(2.8)$
penev_A_F00333	013	—	$\phi^k - \phi$	$\phi^k - \phi$
penev_A_F00334	013	—	ϕ^k	ϕ^k
penev_A_F00335	013	—	$\sim 0.5 \, \%$	$\sim 0.5\%$
penev_A_F00336	013	—	ϵ	ϵ
penev_A_F00337	013	—	$V \approx$	$V \approx$
penev_A_F00338	013	—	$N \approx 400$	$N \approx 400$
penev_A_F00339	014	—	$(T=3279, V=3840)$	$(T = 3279, V = 3840)$
penev_A_F00340	014	—	$(p, q) \in T \times T$	$(p, q) \in T \times T$
penev_A_F00341	014	—	$q = p$	$q = p$
penev_A_F00342	014	—	$q \neq p$	$q \neq p$
penev_A_F00343	014	—	(p, q)	(p, q)
penev_A_F00344	014	—	C_{same}	C_{same}
penev_A_F00345	014	—	C_{diff}	C_{diff}
penev_A_F00346	014	—	$\mu \left(\mathbf{d}^{(p, q)} \right)$	$\mu \left(\mathbf{d}^{(p, q)} \right)$
penev_A_F00347	014	—	$\mathrm{W} \colon S \mapsto S$	$\mathbf{W} \colon S \mapsto S$
penev_A_F00348	014	—	$\mu(\mathbf{d}) = (\mathbf{W} \mathbf{d}, \mathbf{W} \mathbf{d})$	$\mu(\mathbf{d}) = (\mathbf{W} \mathbf{d}, \mathbf{W} \mathbf{d})$
penev_A_F00349	014	—	$\mathbf{d}^{(\mathrm{p}, \mathrm{g})}$	$\mathbf{d}^{(\mathrm{p}, \mathrm{g})}$
penev_A_F00350	014	—	$\mathbf{W}^* \mathbf{W}$	$\mathbf{W}^* \mathbf{W}$
penev_A_F00351	014	—	$\{w_r\}_{r=1}^N$	$\{w_r\}_{r=1}^N$
penev_A_F00352	014	—	$\Phi^* \mathbf{d}^{(p, q)} \in U$	$\Phi^* \mathbf{d}^{(p, q)} \in U$
penev_A_F00353	014	—	$\varepsilon = \left\ \Phi^* \mathbf{d}^{(p, q)} \right\ _U$	$\varepsilon = \left\ \Phi^* \mathbf{d}^{(p, q)} \right\ _U$
penev_A_F00354	014	—	n	n
penev_A_F00355		—	$s(n)$	$s(n)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00356		—	$\frac{n(n-1)}{2}$	$\frac{n(n-1)}{2}$
penev_A_F00357		—	$s(n) \times \frac{n(n-1)}{2}$	$s(n) \times \frac{n(n-1)}{2}$
penev_A_F00358		—	$N_{\{s\}}=3842$	$N_s = 3842$
penev_A_F00359	014	—	$T_{\{\text{restricted}\}}=3049$	$T_{\text{restricted}} = 3049$
penev_A_F00360	014	—	$\{\alpha\}=A$	$\{\alpha\} = A$
penev_A_F00361	014	—	$\alpha \in A$	$\alpha \in A$
penev_A_F00362	014	—	$\alpha \subset T$	$\alpha \subset T$
penev_A_F00363	014	—	$T=\bigcup_A \alpha$	$T = \bigcup_A \alpha$
penev_A_F00364	014	—	$\alpha, \mid \alpha$	$\alpha, \mid \alpha$
penev_A_F00365	014	—	$(p, q) \in C_{\{\text{same}\}}$	$(p, q) \in C_{\text{same}}$
penev_A_F00366	014	—	α	α
penev_A_F00367	014	—	$\frac{ \alpha \times (\alpha \mid -1)}{2}$	$\frac{ \alpha \times (\alpha \mid -1)}{2}$
penev_A_F00368	014	—	$s(n)=\left \left \left\{\left\{\left.\alpha\right _{\left \alpha\right =n}\right\}\right\}\right $	$s(n) = \left \left\{\alpha_{ \alpha =n}\right\}\right $
penev_A_F00369	014	—	$\left \left\{(p, q) \in C_{\{\text{same}\}}\right\}\right , N_{\{s\}}=\frac{1}{2}\left C_{\{\text{same}\}}\right =$	$\left\{(p, q) \in C_{\text{same}}\right\}, N_s = \frac{1}{2}\left C_{\text{same}}\right =$
penev_A_F00370	014	—	$\sum_n s(n) \times \frac{n(n-1)}{2}$	$\sum_n s(n) \times \frac{n(n-1)}{2}$
penev_A_F00371	014	—	$N_{\{d\}}=$	$N_d =$
penev_A_F00372	014	—	$C_{\{s \text{ a m e}\}}$	C_{same}
penev_A_F00373	014	—	$w_{\{r\}}=\sigma_{\{\mathrm{r}\}}\{ \}^2$	$w_r = \sigma_{\mathbf{r}}^2$
penev_A_F00374	014	—	$w_{\{r\}}=1$	$w_r = 1$
penev_A_F00375	014	—	$w_{\{r\}}$	w_r
penev_A_F00376	014	—	$D_{\{r\}}^{\{(p, q)\}}$	$D_r^{(p,q)}$
penev_A_F00377	014	—	$m_{\{r\}}^{\{\text{aame}\}}$	m_r^{aame}
penev_A_F00378	014	—	$m_{\{r\}}^{\{\text{diff}\}}$	m_r^{diff}
penev_A_F00379	014	—	$\sigma_{\{r\}}^{\{\text{an e}\}}$	$\sigma_r^{\text{an e}}$
penev_A_F00380	014	—	$\sigma_{\{r\}}^{\{\text{diff}\}}$	σ_r^{diff}
penev_A_F00381	014	—	$m_{\{r\}}^{\{s \text{ a m e}\}}$	m_r^{same}
penev_A_F00382	014	—	$F_{\{r\}}^{\{s \text{ a m e}\}}(3.6)$	$F_r^{\text{same}}(3.6)$
penev_A_F00383	014	—	$F_{\{r\}}^{\{\text{diff}\}}$	F_r^{diff}
penev_A_F00384	014	—	$\left(m_{\{r\}}^{\{\text{diff}\}} / \sigma_{\{r\}}^{\{\text{dif}\}}\right)$	$\left(m_r^{\text{diff}} / \sigma_r^{\text{dif}}\right)$
penev_A_F00385	014	—	$\left(m_{\{r\}}^{\{\text{same}\}} / \sigma_{\{r\}}^{\{\text{same}\}}\right)$	$\left(m_r^{\text{same}} / \sigma_r^{\text{same}}\right)$
penev_A_F00386	014	—	$\sqrt{2} / 2 \approx 0.707$	$\sqrt{2} / 2 \approx 0.707$
penev_A_F00387	014	—	$(p, q) \in C_{\{\text{diff}\}}$	$(p, q) \in C_{\text{diff}}$
penev_A_F00388	014	—	ϕ^p	ϕ^p
penev_A_F00389	014	—	ϕ^q	ϕ^q
penev_A_F00390	014	—	$\mathcal{N}(0,1)$	$\mathcal{N}(0,1)$
penev_A_F00391	014	—	$\mathcal{N}(0,2)$	$\mathcal{N}(0,2)$
penev_A_F00392	014	—	$m_{\{r\}}^{\{\text{diff}\}}=m_{\{1\}}$	$m_r^{\text{diff}} = m_1$
penev_A_F00393	014	—	$\mathcal{N}(0,2), m_{\{1\}}=2$	$\mathcal{N}(0,2), m_1 = 2$
penev_A_F00394	014	—	$m_{\{2\}}$	m_2
penev_A_F00395	014	—	$\mu_{\{4\}}$	μ_4
penev_A_F00396	014	—	$\mathcal{N}(0,2), m_{\{2\}}=$	$\mathcal{N}(0,2), m_2 =$
penev_A_F00397	014	—	$3 \times 2^2=12$	$3 \times 2^2 = 12$
penev_A_F00398	014	—	$\sigma_{\{r\}}^{\{\text{diff}\}}=\sqrt{m_{\{2\}}-m_{\{1\}}\{ \}^2}=2 \sqrt{2}$	$\sigma_r^{\text{diff}} = \sqrt{m_2 - m_1^2} = 2\sqrt{2}$
penev_A_F00399	014	—	$m_{\{r\}}^{\{\text{d i f}\}} / \sigma_{\{r\}}^{\{\text{d i f}\}}$	$m_r^{\text{dif}} / \sigma_r^{\text{dif}}$
penev_A_F00400	014	—	$\sqrt{2} / 2$	$\sqrt{2} / 2$
penev_A_F00401	015	—	$F_{\{r\}}^{\{s \text{ a m e}\}}$	F_r^{same}
penev_A_F00402		—	$F_{\{r\}}^{\{\text{d i f f}\}}$	F_r^{diff}
penev_A_F00403	014	—	$m_{\{r\}}^{\{\text{diff}\}} / \sigma_{\{r\}}^{\{\text{diff}\}}$	$m_r^{\text{diff}} / \sigma_r^{\text{diff}}$
penev_A_F00404	014	—	$\langle \mathbf{u} \mathbf{v} \rangle \langle \mathbf{u} \mathbf{v} \rangle = \langle \mathbf{u} \rangle^2 \neq 0$	$\langle \mathbf{u} \mathbf{v} \rangle = \langle \mathbf{u} \rangle \langle \mathbf{v} \rangle = \langle \mathbf{u} \rangle^2 \neq 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00405	015	—	$\mu_2 \neq 2$	$\mu_2 \neq 2$
penev_A_F00406	015	—	$N \approx 1800$	$N \approx 1800$
penev_A_F00407	015	—	$D_r^{\{p,\,q\}}(3.5)$	$D_r^{(p,q)}(3.5)$
penev_A_F00408	015	—	ϕ^{p}	$\phi^{\mathfrak{p}}$
penev_A_F00409	015	—	$m_{\mathrm{r}}^{\mathrm{\{ane\}}} / \sigma_{\mathrm{r}}^{\mathrm{\{anme\}}}$	$m_{\mathrm{r}}^{\mathrm{ane}} / \sigma_{\mathrm{r}}^{\mathrm{anme}}$
penev_A_F00410	015	—	$m_{\mathrm{r}}^{\mathrm{\{diff\}}} / \sigma_{\mathrm{r}}^{\mathrm{\{diff\}}}$	$m_{\mathrm{r}}^{\mathrm{diff}} / \sigma_{\mathrm{r}}^{\mathrm{diff}}$
penev_A_F00411	015	—	$F_r^{\mathrm{\{same\}}}$	F_r^{same}
penev_A_F00412	015	—	$\langle \mathbf{u} \mathbf{v} \rangle \neq 0$	$\langle \mathbf{uv} \rangle \neq 0$
penev_A_F00413	015	—	$T \times T$	$T \times T$
penev_A_F00414	015	—	$\frac{1}{2} C_{\mathrm{diff}} = N_d = 4,642,834$	$\frac{1}{2} C_{\mathrm{diff}} = N_d = 4,642,834$
penev_A_F00415	015	—	$\frac{T \times T}{2} = \frac{3049 \times 3048}{2} = 4,646,676$	$\frac{T \times T}{2} = \frac{3049 \times 3048}{2} = 4,646,676$
penev_A_F00416	015	—	u	u
penev_A_F00417	015	—	$F: \mathcal{R} \mapsto [0,1]$	$F: \mathcal{R} \mapsto [0,1]$
penev_A_F00418	015	—	$F(u)$	$F(u)$
penev_A_F00419	015	—	$P\{\mathbf{u} < u\}$	$P\{\mathbf{u} < u\}$
penev_A_F00420	015	—	F_1	F_1
penev_A_F00421	015	—	F_2	F_2
penev_A_F00422	015	—	$F_2 \circ F_1^{-1}: [0,1] \mapsto [0,1]$	$F_2 \circ F_1^{-1}: [0,1] \mapsto [0,1]$
penev_A_F00423	015	—	$r=2$	$r = 2$
penev_A_F00424	015	—	$D_2^{\{p,\,q\}}$	$D_2^{(p,q)}$
penev_A_F00425	015	—	30%	30%
penev_A_F00426	015	—	$F_2^{\mathrm{\{same\}}}$	F_2^{same}
penev_A_F00427	015	—	$F_2^{\mathrm{\{dif\}}}$	F_2^{dif}
penev_A_F00428	015	—	80%	80%
penev_A_F00429	015	—	$w_{\tau}^{\mathrm{\{equal\}}} = 1$	$w_{\tau}^{\mathrm{equal}} = 1$
penev_A_F00430	015	—	D_r	D_r
penev_A_F00431	015	—	$\mathrm{d}^{\{(\mathrm{p},\,q)\}}$	$\mathbf{d}^{(\mathfrak{p},q)}$
penev_A_F00432		—	(3.8)	(3.8)
penev_A_F00433	015	—	$x=F_1(u),\,y=F_2(u)$	$x = F_1(u), y = F_2(u)$
penev_A_F00434	015	—	$w_r^{\mathrm{\{Fieher\}}}(3.8)$	$w_r^{\mathrm{Fieher}}(3.8)$
penev_A_F00435	015	—	$H^{\mathrm{\{intra\}}}$	H^{intra}
penev_A_F00436	015	—	$H_{rr}^{\mathrm{\{intra\}}} = \left(\sigma_r^{\mathrm{\{same\}}}\right)^2 + \left(\sigma_r^{\mathrm{\{diff\}}}\right)^2$	$H_{rr}^{\mathrm{intra}} = \left(\sigma_r^{\mathrm{same}}\right)^2 + \left(\sigma_r^{\mathrm{diff}}\right)^2$
penev_A_F00437	015	—	$w_r^{\mathrm{\{Fisher\}}}, w_r^{\mathrm{\{nnive\}}}$	$w_r^{\mathrm{Fisher}}, w_r^{\mathrm{nnive}}$
penev_A_F00438	015	—	$w_r^{\mathrm{\{equal\}}}$	w_r^{equal}
penev_A_F00439	015	—	$w_r^{\mathrm{\{Fisher\}}}$	w_r^{Fisher}
penev_A_F00440	015	—	$N=20$	$N = 20$
penev_A_F00441	015	—	$D^{\mathrm{\{same\}}}$	D^{same}
penev_A_F00442	015	—	$D^{\mathrm{\{diff\}}}$	D^{diff}
penev_A_F00443	015	—	ε	ε
penev_A_F00444	015	—	20%	20%
penev_A_F00445	015	—	$w_r^{\mathrm{\{naive\}}}(3.9)$	$w_r^{\mathrm{naive}}(3.9)$
penev_A_F00446	016	—	$\left\{\Psi_r(x)\right\}_{r=1}^{16}$	$\{\Psi_r(x)\}_{r=1}^{16}$
penev_A_F00447	016	—	$(T=1038,\,V=3840)$	$(T = 1038, V = 3840)$
penev_A_F00448	016	—	$\left\{\phi^t(\mathbf{x})\right\},\,t=1,\,\cdots,\,T$	$\{\phi^t(\mathbf{x}), t = 1, \cdots, T\}$
penev_A_F00449	017	—	$V \approx 6 \times 10^6, R(\mathbf{x}, \mathbf{y})$	$V \approx 6 \times 10^6, R(\mathbf{x}, \mathbf{y})$
penev_A_F00450	017	—	$\frac{1}{2} V^2 \approx 18,000$	$\frac{1}{2} V^2 \approx 18,000$
penev_A_F00451	017	—	$\sim 6,000,000$	$\sim 6,000,000$
penev_A_F00452		—	$r(\Delta \mathbf{x})$	$r(\Delta \mathbf{x})$
penev_A_F00453	017	—	$\ell = \ \mathbf{x} - \mathbf{y}\ $	$\ell = \ \mathbf{x} - \mathbf{y}\ $
penev_A_F00454	017	—	$\mathbf{x} - \mathbf{y}$	$\mathbf{x} - \mathbf{y}$
penev_A_F00455	017	—	$\sim \sqrt{6,000,000} \approx 3,000$	$\sim \sqrt{6,000,000} \approx 3,000$
penev_A_F00456	017	—	$\mathbf{f}$	$\mathbf{f}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00457	017	—	$\Psi_{\mathbf{f}}(\mathbf{x})$	$\Psi_{\mathbf{f}}(\mathbf{x})$
penev_A_F00458	017	—	$r(\mathbf{x})$	$r(\mathbf{x})$
penev_A_F00459	017	—	$\mathbf{f}, \tilde{r}(\mathbf{f})$	$\mathbf{f}, \tilde{r}(\mathbf{f})$
penev_A_F00460	017	—	$ f $	$ f $
penev_A_F00461	017	—	$\rho(\ell)$	$\rho(\ell)$
penev_A_F00462	017	—	$\alpha \approx 2$	$\alpha \approx 2$
penev_A_F00463	017	—	ℓ	ℓ
penev_A_F00464	017	—	$\alpha = 2$	$\alpha = 2$
penev_A_F00465	017	—	$(2.25)^9$	$(2.25)^9$
penev_A_F00466	017	—	$\Psi_{\mathbf{f}}$	$\Psi_{\mathbf{f}}$
penev_A_F00467	017	—	$\sigma_{\mathbf{f}}$	$\sigma_{\mathbf{f}}$
penev_A_F00468	017	—	$N=V \approx 6,000,000$	$N = V \approx 6,000,000$
penev_A_F00469	017	—	$N \approx 6,000,000$	$N \approx 6,000,000$
penev_A_F00470	017	—	$N \times V$	$N \times V$
penev_A_F00471	017	—	$N=V$	$N = V$
penev_A_F00472		—	$1,000,000$	$1,000,000$
penev_A_F00473	019	—	σ_{f}	σ_{f}
penev_A_F00474	018	—	$N=6,000,000$	$N = 6,000,000$
penev_A_F00475	018	—	$\phi^{\hat{t}} = \Phi^* \hat{t}$	$\phi^{\hat{t}} = \Phi^* \hat{t}$
penev_A_F00476	018	—	$\Phi, \mathbf{K}$	$\Phi, \mathbf{K}$
penev_A_F00477	018	—	$O(\mathbf{x})$	$O(\mathbf{x})$
penev_A_F00478	018	—	O^*O	O^*O
penev_A_F00479	018	—	$\sigma_{\mathrm{f}} > 0$	$\sigma_{\mathrm{f}} > 0$
penev_A_F00480	018	—	$\Psi_{\mathbf{F}}$	$\Psi_{\mathbf{F}}$
penev_A_F00481	018	—	$\{\mathbf{o}_{\mathrm{f}}\}$	$\{\mathbf{o}_{\mathrm{f}}\}$
penev_A_F00482	018	—	$\mathbf{o}_{\mathrm{f}}^{14}$	$\mathbf{o}_{\mathrm{f}}^{14}$
penev_A_F00483	018	—	$O = O^{*+}$	$O = O^{*+}$
penev_A_F00484	018	—	1^{15}	1^{15}
penev_A_F00485	018	—	$U(4.12), O O^*$	$U(4.12), O O^*$
penev_A_F00486	018	—	$T \gg V^{16}$	$T \gg V^{16}$
penev_A_F00487	018	—	$\{\mathbf{o}_{\mathrm{f}}\}$	$\{\mathbf{o}_{\mathrm{f}}\}$
penev_A_F00488	018	—	$\{\mathbf{s}_{\mathrm{f}}\}$	$\{\mathbf{s}_{\mathrm{f}}\}$
penev_A_F00489	018	—	$\mathbf{U}^* \mathbf{U} = \mathbf{P}'$	$\mathbf{U}^* \mathbf{U} = \mathbf{P}'$
penev_A_F00490	018	—	$\mathbf{U} = \mathbf{P}'$	$\mathbf{U} = \mathbf{P}'$
penev_A_F00491	018	—	$V = T$	$V = T$
penev_A_F00492	018	—	$1/\sigma_{\mathrm{f}}$	$1/\sigma_{\mathrm{f}}$
penev_A_F00493	018	—	$\Phi^{*+} \phi \in S$	$\Phi^{*+} \phi \in S$
penev_A_F00494	018	—	$\phi, -\log \mathcal{P}[\phi]$	$\phi, -\log \mathcal{P}[\phi]$
penev_A_F00495	018	—	$O = \mathbb{K} \phi$	$O = \mathbb{K} \phi$
penev_A_F00496	018	—	$\ \Phi^{*+} \phi\ _S^2$	$\ \Phi^{*+} \phi\ _S^2$
penev_A_F00497	018	—	$\{O(\mathbf{x})\}_{x \in V}$	$\{O(\mathbf{x})\}_{x \in V}$
penev_A_F00498	018	—	$\mathbb{K}(4.19)^{18}$	$\mathbb{K}(4.19)^{18}$
penev_A_F00499	018	—	$O = \mathbb{K} \phi, O^{*+} = O$	$O = \mathbb{K} \phi, O^{*+} = O$
penev_A_F00500	018	—	$\Phi^* O^{*+} O = \Phi^* \Phi^{*+} \phi = \phi$	$\Phi^* O^{*+} O = \Phi^* \Phi^{*+} \phi = \phi$
penev_A_F00501		—	$1/\sigma_r$	$1/\sigma_r$
penev_A_F00502		—	σ	σ
penev_A_F00503		—	τ	τ
penev_A_F00504		—	$(4, 24)$	$(4, 24)$
penev_A_F00505	019	—	$^{19}\Psi_{\mathrm{f}}$	$^{19}\Psi_{\mathrm{f}}$
penev_A_F00506	019	—	$\Psi_{\mathbf{f}}(\mathbf{x}) = (\Psi_{\mathbf{f}}, \hat{\mathbf{x}})$	$\Psi_{\mathbf{f}}(\mathbf{x}) = (\Psi_{\mathbf{f}}, \hat{\mathbf{x}})$
penev_A_F00507	019	—	$\mathbf{x}_0 = \mathbf{0}$	$\mathbf{x}_0 = \mathbf{0}$
penev_A_F00508	019	—	$\mathbf{x}_0$	$\mathbf{x}_0$
penev_A_F00509	019	—	$\mathbb{K}_{\mathbf{x}}$	$\mathbb{K}_{\mathbf{x}}$
penev_A_F00510	019	—	$\mathbf{y}, \mathbb{K}_{\mathbf{x}}(\mathbf{y}) = (\hat{\mathbf{y}}, \mathbb{K}_{\mathbf{x}})$	$\mathbf{y}, \mathbb{K}_{\mathbf{x}}(\mathbf{y}) = (\hat{\mathbf{y}}, \mathbb{K}_{\mathbf{x}})$
penev_A_F00511	019	—	$i8^{21}$	$i8^{21}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00512	019	—	$\mathrm{\mathbf{K}}_{\mathrm{\mathbf{x}}}$	$\mathbf{K}_{\mathbf{x}}$
penev_A_F00513	019	—	K^{n}	K^n
penev_A_F00514	019	—	K^{0}	K^0
penev_A_F00515	019	—	$\mathrm{\mathbf{R}}=\mathrm{\mathbf{K}}^{\mathrm{-2}}=$	$\mathbf{R} = \mathbf{K}^{-2} =$
penev_A_F00516	019	—	$r(\mathrm{\mathbf{x}}-\mathrm{\mathbf{y}})$	$r(\mathbf{x} - \mathbf{y})$
penev_A_F00517	019	—	$\mathrm{\mathbf{x}}-\mathrm{\mathbf{y}}=0$	$\mathbf{x} - \mathbf{y} = 0$
penev_A_F00518	019	—	$\mathrm{\mathbf{K}}^{\mathrm{0}}=\mathrm{\mathbf{1}}$	$\mathbf{K}^0 = \mathbf{1}$
penev_A_F00519	019	—	K^{1}	K^1
penev_A_F00520	019	—	$\mathrm{\mathbf{K}}^{\mathrm{-2}}$	$\mathbf{K}^{-2}$
penev_A_F00521	019	—	$\mathrm{\mathbf{K}}^{\mathrm{0}}$	$\mathbf{K}^0$
penev_A_F00522	019	—	$\mathrm{\mathbf{K}}^{\mathrm{1}}$	$\mathbf{K}^1$
penev_A_F00523	019	—	K^{0}	K^0
penev_A_F00524	019	—	0	O
penev_A_F00525		—	$\{ \ }^{\mathrm{21}}\mathrm{\mathbf{K}}^{\mathrm{2}}\equiv\mathrm{\mathbf{R}}^{\mathrm{-1}}$	$^{21}\mathbf{K}^2 \equiv \mathbf{R}^{-1}$
penev_A_F00526		—	$\Phi^{\mathrm{+}}=\sum_{\mathrm{\mathbf{f}}}\mathrm{\Psi}_{\mathrm{\mathbf{f}}}\frac{1}{\sigma_{\mathrm{\mathbf{f}}}}\left(\mathrm{\mathbf{\Xi}}_{\mathrm{\mathbf{f}}},\right)$	<i>(not rendered)</i>
penev_A_F00527		—	$\mathrm{\mathbf{K}}^{\mathrm{2}}$	$\mathbf{K}^2$
penev_A_F00528	019	—	$\mathrm{K}^{\mathrm{-1}}$	K^{-1}
penev_A_F00529	019	—	$\mathrm{\mathbf{0}}^{\mathrm{*}}\mathrm{\mathbf{0}}=\mathrm{\mathbf{1}}_{\mathrm{U}}$	$\mathbf{0}^*\mathbf{0} = \mathbf{1}_U$
penev_A_F00530	019	—	$\mathrm{\mathbf{1}}_{\mathrm{U}}$	$\mathbf{1}_U$
penev_A_F00531	020	—	$\{O(\mathrm{\mathbf{x}})\}$	$\{O(\mathbf{x})\}$
penev_A_F00532	020	—	a_{r}	a_r
penev_A_F00533	020	—	$\left\{a_{\mathrm{r}}\right\}$	$\{a_r\}$
penev_A_F00534	020	—	$\mathrm{\mathbf{K}}^{\mathrm{(-1)}}$	$\mathbf{K}^{(-1)}$
penev_A_F00535		—	(4.35)	(4.35)
penev_A_F00536	020	—	$\left \left \phi-\phi^{\mathrm{rec}}\right \right ^2$	$\ \phi - \phi^{\mathrm{rec}}\ ^2$
penev_A_F00537	020	—	$\left \left \phi_{\perp}(\mathrm{\mathbf{x}})\right \right ^2$	$\ \phi_{\perp}(\mathbf{x})\ ^2$
penev_A_F00538	020	—	$\phi_{\perp}(\mathrm{\mathbf{x}})$	$\phi_{\perp}(\mathbf{x})$
penev_A_F00539	020	—	$\mathrm{\mathbf{x}}_{\mathrm{0}}$	$\mathbf{x}_0$
penev_A_F00540	020	—	$a\,e$	$a e$
penev_A_F00541	020	—	$a-e$	$a - e$
penev_A_F00542	020	—	$K\left(\mathrm{\mathbf{x}}_{\mathrm{0}},\mathrm{\mathbf{y}}\right)=\mathrm{\mathbf{K}}_{\mathrm{\mathbf{x}}_{\mathrm{0}}}\left(\mathrm{\mathbf{y}}\right)=\left(\mathrm{\mathbf{K}},\hat{\mathrm{\mathbf{x}}}_{\mathrm{0}}\right)$	$K\left(\mathbf{x}_0,\mathbf{y}\right)=\mathbf{K}_{\mathbf{x}_0}(\mathbf{y})=\left(\mathbf{K},\hat{\mathbf{x}}_0\right)$
penev_A_F00543	020	—	$\mathbf{P}_{\mathbf{x}_0}(\mathbf{y})$	$\mathbf{P}_{\mathbf{x}_0}(\mathbf{y})$
penev_A_F00544	020	—	$V=8192,\,T=N=348$	$V=8192,T=N=348$
penev_A_F00545	020	—	$n=0$	$n=0$
penev_A_F00546	020	—	$\theta=0$	$\theta=0$
penev_A_F00547	020	—	1 D	1D
penev_A_F00548	020	—	4.1 c, 4.1 d	4.1c,4.1d
penev_A_F00549	020	—	4.1 e	4.1e
penev_A_F00550	020	—	4.2 a, 4.2 b	4.2a,4.2b
penev_A_F00551	020	—	4.2 c, 4.2 d	4.2c,4.2d
penev_A_F00552	020	—	4.2 e	4.2e
penev_A_F00553	020	—	$\mathrm{\mathbf{x}}_{\mathrm{\mathbf{0}}}$	$\mathbf{x}_0$
penev_A_F00554	020	—	$a\,\,\,\mathrm{e}$	$a\,\,\,e$
penev_A_F00555	020	—	a	a
penev_A_F00556	020	—	$K\left(\mathrm{\mathbf{x}}_{\mathrm{0}},\mathrm{\mathbf{y}}\right)=\mathrm{\mathbf{K}}_{\mathrm{\mathbf{x}}_{\mathrm{0}}}\left(\mathrm{\mathbf{y}}\right)=\left(\mathrm{\mathbf{K}},\hat{\mathrm{\mathbf{x}}}_{\mathrm{0}}\right)$	$K\left(\mathbf{x}_0,\mathbf{y}\right)=\mathbf{K}_{\mathbf{x}_0}(\mathbf{y})=\left(\mathbf{K},\hat{\mathbf{x}}_0\right)$
penev_A_F00557	020	—	$F_{\mathrm{\mathrm{r}}}=\frac{\sigma_{\mathrm{r}}^2}{\left(\sigma_{\mathrm{r}}^2+n^2\right)}\left(\sigma_{\mathrm{r}}^2+n^2\right)$	$F_{\mathrm{r}}=\frac{\sigma_{\mathrm{r}}^2}{\left(\sigma_{\mathrm{r}}^2+n^2\right)}$
penev_A_F00558	020	—	$\mathrm{K}=\sum_{\mathrm{r}=1}^N\psi_{\mathrm{r}}\frac{F_{\mathrm{r}}}{\sigma_{\mathrm{r}}}(\psi_{\mathrm{r}},)$	$\mathrm{K}=\sum_{\mathrm{r}=1}^N\psi_{\mathrm{r}}\frac{F_{\mathrm{r}}}{\sigma_{\mathrm{r}}}(\psi_{\mathrm{r}},)$
penev_A_F00559	020	—	(T=1038, V=3840), N=400	$(T=1038,V=3840),N=400$
penev_A_F00560	020	—	$n=\sigma_{400}$	$n=\sigma_{400}$
penev_A_F00561	020	—	r=400	$r=400$
penev_A_F00562	020	—	1 C	1C
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00563	020	—	$0_{\{N\}}$	O_N
penev_A_F00564	021	—	$0_{\{N\}}=\sum_{r=1}^N a_{\{r\}} F_{\{r, N\}} \psi_{\{r\}}$	$O_N=\sum_{r=1}^N a_r F_{r,N} \psi_r$
penev_A_F00565	021	—	$F_{\{\mathrm{r}\}, N}=\frac{\sigma_{\{\mathrm{r}\}}^2}{\alpha_{\mathrm{r}}^2+n_{\mathrm{N}}^2}\},n_{\{N\}}=\sigma_{\{N\}}$	$F_{\mathrm{r},N}=\frac{\sigma_{\mathrm{r}}^2}{\alpha_{\mathrm{r}}^2+n_{\mathrm{N}}^2},n_N=\sigma_N$
penev_A_F00566	021	—	$5\ x$	$5x$
penev_A_F00567	021	—	$20\ x$	$20x$
penev_A_F00568	021	—	$1\ E$	$1E$
penev_A_F00569	021	—	$N\ \sim 60$	$N\sim 60$
penev_A_F00570	021	—	$N\ \sim 200$	$N\sim 200$
penev_A_F00571	021	—	$N\ \sim 500$	$N\sim 500$
penev_A_F00572	021	—	$N\ \sim 300, 0(\mathrm{x})$	$N\sim 300,O(\mathbf{x})$
penev_A_F00573	021	—	$\{0(\mathbf{x})\}_{\mathbf{x}\in V}$	$\{O(\mathbf{x})\}_{\mathbf{x}\in V}$
penev_A_F00574	021	—	$N\lll V$	$N\ll V$
penev_A_F00575	021	—	$\left\{a_{\{r\}}\right\}_{r\in N}$	$\{a_r\}_{r\in N}$
penev_A_F00576	021	—	$\left\{\mathbf{x}_{\{m\}}\right\}\equiv\mathcal{M},\left\{O_{\{m\}}\equiv O\left(\mathbf{x}_{\{m\}}\right)\right\}_{\mathbf{x}_{\{m\}}\in\mathcal{M}}$	$\{\mathbf{x}_m\}\equiv\mathcal{M},\{O_m\equiv O(\mathbf{x}_m)\}_{\mathbf{x}_m\in\mathcal{M}}$
penev_A_F00577	021	—	$0_{\{m\}}=\sum_{r=1}^N\boldsymbol{\Psi}_{\{r\ m\}} a_{\{r\}}$	$O_m=\sum_{r=1}^N\Psi_{rm}a_r$
penev_A_F00578	021	—	$\boldsymbol{\Psi}_{\{r\ m\}}\equiv\Psi_{\{r\}}\left(\mathbf{x}_{\{m\}}\right)$	$\Psi_{rm}\equiv\Psi_r(\mathbf{x}_m)$
penev_A_F00579	021	—	$N\,\times \mathcal{M} $	$N\times \mathcal{M} $
penev_A_F00580	021	—	$ \mathcal{M} =N,\ \Psi$	$ \mathcal{M} =N,\Psi$
penev_A_F00581	021	—	$a_{\{r\}}=\sum_{\{m\}}0_{\{m\}}\ \Psi^{\{-1\}}_{\{m\ r\}}$	$a_r=\sum_mO_m\Psi^{-1}_{mr}$
penev_A_F00582	021	—	$a_{\{m\}}(\mathbf{x})=\sum_{r=1}^N\Psi_{\{m\ r\}}^{\{-1\}}\Psi_{\{r\}}(\mathbf{x})$	$a_m(\mathbf{x})=\sum_{r=1}^N\Psi_{mr}^{-1}\Psi_r(\mathbf{x})$
penev_A_F00583	021	—	$\left\{a_{\{m\}}(\mathbf{x})\right\}_{\mathbf{x}\in\mathcal{M}}$	$\{a_m(\mathbf{x})\}_{\mathbf{x}_m\in\mathcal{M}}$
penev_A_F00584	021	—	$\mathcal{M}$	$\mathcal{M}$
penev_A_F00585	021	—	$ \mathcal{M} $	$ \mathcal{M} $
penev_A_F00586	021	—	$\mathbf{P}_{\{\mathbf{x}\}}$	$\mathbf{P}_{\mathbf{x}}$
penev_A_F00587	021	—	$\left\{O\left(\mathbf{x}_{\{m\}}\right)\right\}_{x_{\{m\}}\in\mathcal{M}}$	$\{O(\mathbf{x}_m)\}_{x_m\in\mathcal{M}}$
penev_A_F00588	021	—	$0_{\{m\}}$	O_m
penev_A_F00589	021	—	$V\text{-}N\gg N$	$V-N\gg N$
penev_A_F00590	021	—	$O(x)$	$O(x)$
penev_A_F00591	021	—	$\mathbf{x}_{\{m\}}$	$\mathbf{x}_m$
penev_A_F00592	021	—	$\mathbf{K}_{\{\mathbf{x}_{\{m\}}\}}$	$\mathbf{K}_{\mathbf{x}_m}$
penev_A_F00593	021	—	$\mathbf{P}_{\{x_{\{m\}}\}}$	$\mathbf{P}_{x_m}$
penev_A_F00594	021	—	$O\left(\mathbf{x}_{\{m\}}\right)$	$O(\mathbf{x}_m)$
penev_A_F00595	021	—	$\left\{O\left(\mathbf{x}_{\mathbf{m}}\right)\right\}_{\mathbf{x}_{\mathbf{m}}\in\mathcal{M}}$	$\{O(\mathbf{x}_{\mathbf{m}})\}_{\mathbf{x}_{\mathbf{m}}\in\mathcal{M}}$
penev_A_F00596	021	—	$\left\{O\left(\mathbf{x}_{\{m\}}\right)\right\}_{\mathbf{x}_{\{m\}}\in\mathcal{M}}$	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m\in\mathcal{M}}$
penev_A_F00597	021	—	$\phi=\Phi^{\{*\}}\ \mathrm{s}$	$\phi=\Phi^*\ \mathbf{s}$
penev_A_F00598	021	—	$0=\mathrm{0}^{\{*\}}$	$O=\mathbf{O}^*$
penev_A_F00599	021	—	$0_{\{m\}}=0\left(\mathrm{x}_{\{m\}}\right)$	$O_m=O(\mathbf{x}_m)$
penev_A_F00600	021	—	$0^{\{\text{rec}\}}(\mathbf{x})$	$O^{\mathrm{rec}}(\mathbf{x})$
penev_A_F00601	021	—	$\mathcal{P}[\phi(\mathbf{x})](2.2)$	$\mathcal{P}[\phi(\mathbf{x})](2.2)$
penev_A_F00602	021	—	$0_{\{m\}}\,?^{29}$	$O_m?^{29}$
penev_A_F00603	021	—	$0_{\{m\}}=0\left(\mathbf{x}_{\{m\}}\right)=\left(\hat{\mathbf{x}}_{\{m\}},\ 0\right)=\left(\hat{\mathbf{x}}_{\{m\}},\ \mathbf{0}\ast\ \mathrm{s}\right)=\left(\mathbf{0}\hat{\mathbf{x}}_{\{m\}},\ s\right)$	$O_m=O(\mathbf{x}_m)=(\hat{\mathbf{x}}_m,O)=(\hat{\mathbf{x}}_m,\mathbf{O}\ast\ \mathbf{s})=\langle\mathbf{O}\hat{\mathbf{x}}_m,s\rangle$
penev_A_F00604	021	—	$\mathrm{q}_{\{\mathrm{x}\}}\in S$	$\mathbf{q}_{\mathbf{x}}\in S$
penev_A_F00605	021	—	$0(\mathrm{x})$	$O(\mathbf{x})$
penev_A_F00606	022	—	$\mathrm{x}_{\{\mathrm{m}\}}$	$\mathbf{x}_{\mathbf{m}}$
penev_A_F00607	022	—	$\mathrm{q}_{\{x\}}^{\{30\}}$	$\mathbf{q}_x^{30}$
penev_A_F00608	022	—	$\mathbf{q}_{\{\mathbf{x}_{\{m\}}\}}$	$\mathbf{q}_{\mathbf{x}_m}$
penev_A_F00609	022	—	$\mathbf{s}=\mathbf{s}_{\parallel}+\mathbf{s}_{\perp}$	$\mathbf{s}=\mathbf{s}_{\parallel}+\mathbf{s}_{\perp}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 ■ no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00610	022	—	$\mathbf{Q}_{\mathbf{0}_{\mathbf{x}_{\mathbf{m}}}}^{\mathbf{p}_{\mathbf{x}_{\mathbf{m}}}}=\mathbf{p}_{\mathbf{x}_{\mathbf{m}}}\left\langle \mathbf{q}_{\mathbf{x}_{\mathbf{m}}},\mathbf{p}_{\mathbf{x}_{\mathbf{m}}}\right\rangle \left\langle \mathbf{q}_{\mathbf{x}_{\mathbf{m}}},\right\rangle =\mathbf{p}_{\mathbf{x}_{\mathbf{m}}}\left\langle \mathbf{q}_{\mathbf{x}_{\mathbf{m}}},\right\rangle =\mathbf{Q}_{\mathbf{x}_{\mathbf{m}}};\mathbf{Q}_{\mathbf{x}_{\mathbf{m}}}$	$\mathbf{Q}_{\mathbf{x}_m}^2=\mathbf{p}_{\mathbf{x}_m}\left\langle \mathbf{q}_{\mathbf{x}_m},\mathbf{p}_{\mathbf{x}_m}\right\rangle \left\langle \mathbf{q}_{\mathbf{x}_m},\right\rangle =\mathbf{p}_{\mathbf{x}_m}\left\langle \mathbf{q}_{\mathbf{x}_m},\right\rangle =\mathbf{Q}_{\mathbf{x}_m};\mathbf{Q}_{\mathbf{x}_m}$
penev_A_F00611	022	—	$\left\langle \mathbf{q}_{\mathbf{x}_{\mathbf{m}}},\mathbf{s}_{\parallel }\right\rangle =O_m$	$\left\langle \mathbf{q}_{\mathbf{x}_m},\mathbf{s}_{\parallel }\right\rangle =O_m$
penev_A_F00612	022	—	$\mathbf{s}_{\parallel }=\alpha \mathbf{q}_{x_m}$	$\mathbf{s}_{\parallel }=\alpha \mathbf{q}_{x_m}$
penev_A_F00613	022	—	$\alpha =\frac{O_m}{\left\langle \mathbf{q}_{x_m},\mathbf{q}_{x_m}\right\rangle }$	$\alpha =\frac{O_m}{\left\langle \mathbf{q}_{x_m},\mathbf{q}_{x_m}\right\rangle }$
penev_A_F00614	022	—	$\left\langle \mathbf{s}_{\parallel },\mathbf{s}_{\perp }\right\rangle =0$	$\left\langle \mathbf{s}_{\parallel },\mathbf{s}_{\perp }\right\rangle =0$
penev_A_F00615	022	—	$\mathcal{P}[\phi]\propto \exp \left\{ -\frac{1}{2}\left\ \mathbf{s}\right\ ^2\right\} \left(2.16\right)$	$\mathcal{P}[\phi]\propto \exp \left\{ -\frac{1}{2}\left\ \mathbf{s}\right\ ^2\right\} \left(2.16\right)$
penev_A_F00616	022	—	$\left\ \mathbf{s}_{\perp }\right\ ^2=0\left(\mathbf{s}_{\perp }=0\right)$	$\left\ \mathbf{s}_{\perp }\right\ ^2=0\left(\mathbf{s}_{\perp }=0\right)$
penev_A_F00617	022	—	$O_m,\mathbf{s}^{\mathrm{rec}}=\mathbf{s}_{\parallel }$	$O_m,\mathbf{s}^{\mathrm{rec}}=\mathbf{s}_{\parallel }$
penev_A_F00618	022	—	$\mathbf{q}_{\mathbf{x}_m}\in S$	$\mathbf{q}_{\mathbf{x}_m}\in S$
penev_A_F00619	022	—	$U_{\mathbf{x}_m}\subset S$	$U_{\mathbf{x}_m}\subset S$
penev_A_F00620	022	—	$\mathbf{Q}_{\mathbf{x}_m}$	$\mathbf{Q}_{\mathbf{x}_m}$
penev_A_F00621	022	—	$\mathbf{s}_{\parallel }\in U_{\mathbf{x}_m}$	$\mathbf{s}_{\parallel }\in U_{\mathbf{x}_m}$
penev_A_F00622	022	—	$\mathbf{s}_{\perp }\in U_{\mathbf{x}_m}^1$	$\mathbf{s}_{\perp }\in U_{\mathbf{x}_m}^1$
penev_A_F00623	022	—	$\mathbf{s}_{\parallel }$	$\mathbf{s}_{\parallel }$
penev_A_F00624	022	—	$\mathbf{p}_{\mathbf{x}_m}$	$\mathbf{p}_{\mathbf{x}_m}$
penev_A_F00625	022	—	$\mathbf{s}_{\perp }\in U_{x_m}^{\perp }$	$\mathbf{s}_{\perp }\in U_{x_m}^{\perp }$
penev_A_F00626	022	—	$\mathbf{s}^{\mathrm{rec}}\in S$	$\mathbf{s}^{\mathrm{rec}}\in S$
penev_A_F00627	022	—	$\mathbf{s}_{\perp }=0$	$\mathbf{s}_{\perp }=0$
penev_A_F00628	022	—	$\left\{ \mathbf{q}_{\mathbf{x}_m}\right\} _{\mathbf{x}_m\in \mathcal{M}}$	$\left\{ \mathbf{q}_{\mathbf{x}_m}\right\} _{\mathbf{x}_m\in \mathcal{M}}$
penev_A_F00629	022	—	$\left\{ \mathbf{o}_{\mathbf{r}}\right\} _{r\in N}$	$\left\{ \mathbf{o}_{\mathbf{r}}\right\} _{r\in N}$
penev_A_F00630	022	—	$P_{\mathbf{x}_m}(\mathbf{x})$	$P_{\mathbf{x}_m}(\mathbf{x})$
penev_A_F00631	022	—	$\mathbf{x}_m$	$\mathbf{x}_m$
penev_A_F00632	022	—	$\left\langle \mathbf{q}_{\mathbf{x}_m},\mathbf{q}_{\mathbf{x}_m}\right\rangle =\left\langle \mathbf{O}\hat{\mathbf{x}}_m,\mathbf{q}_{\mathbf{x}_m}\right\rangle =\left(\hat{\mathbf{x}}_m,\mathbf{O}^*\mathbf{q}_{\mathbf{x}_m}\right)=P_{\mathbf{x}_m}\left(\mathbf{x}_m\right);$	$\left\langle \mathbf{q}_{\mathbf{x}_m},\mathbf{q}_{\mathbf{x}_m}\right\rangle =\left\langle \mathbf{O}\hat{\mathbf{x}}_m,\mathbf{q}_{\mathbf{x}_m}\right\rangle =\left(\hat{\mathbf{x}}_m,\mathbf{O}^*\mathbf{q}_{\mathbf{x}_m}\right)=P_{\mathbf{x}_m}\left(\mathbf{x}_m\right);$
penev_A_F00633	022	—	$a_{\mathbf{x}_m}(\mathbf{x})$	$a_{\mathbf{x}_m}(\mathbf{x})$
penev_A_F00634	022	—	$a_m(\mathbf{x})$	$a_m(\mathbf{x})$
penev_A_F00635	022	—	$P_{\mathbf{x}_m}(\mathbf{x})$	$P_{\mathbf{x}_m}(\mathbf{x})$
penev_A_F00636	022	—	$\mathbf{x}_m$	$\mathbf{x}_m$
penev_A_F00637	022	—	$\mathbf{x}_m,O^{rec}(\mathbf{x}_m)=O_m$	$\mathbf{x}_m,O^{rec}(\mathbf{x}_m)=O_m$
penev_A_F00638	022	—	$\left\langle \mathbf{q}_{\mathbf{x}_m},\mathbf{s}^{\mathrm{rec}}\right\rangle =O_m$	$\left\langle \mathbf{q}_{\mathbf{x}_m},\mathbf{s}^{\mathrm{rec}}\right\rangle =O_m$
penev_A_F00639	022	—	$K^{(-1)}_{x_m}(x)=\left(\hat{x},K^{(-1)}x_m\right)$	$K^{(-1)}_{x_m}(x)=\left(\hat{x},K^{(-1)}x_m\right)$
penev_A_F00640	022	—	$\phi ^{\mathrm{rec}}(\mathbf{x})$	$\phi ^{\mathrm{rec}}(\mathbf{x})$
penev_A_F00641	022	—	$\mathbf{P}_{\mathbf{x}}$	$\mathbf{P}_{\mathbf{x}}$
penev_A_F00642	022	—	$\mathbf{a}_{\mathbf{l}}$	$\mathbf{a}_{\mathbf{l}}$
penev_A_F00643	022	—	$a_{\mathbf{x}_m}(\mathbf{x})$	$a_{\mathbf{x}_m}(\mathbf{x})$
penev_A_F00644	022	—	$O^{\mathrm{rec}}(\mathbf{x}_m)\neq$	$O^{\mathrm{rec}}(\mathbf{x}_m)\neq$
penev_A_F00645	022	—	$O^{res}(\mathbf{x}_m)$	$O^{res}(\mathbf{x}_m)$
penev_A_F00646	022	—	$a_{\mathbf{x}_m}(\mathbf{x})$	$a_{\mathbf{x}_m}(\mathbf{x})$
penev_A_F00647	022	—	$O(\mathbf{x}_m);O(\mathbf{x})$	$O(\mathbf{x}_m);O(\mathbf{x})$
penev_A_F00648	022	—	$O^{\mathrm{rec}}(\mathbf{x})-O(\mathbf{x})$	$O^{\mathrm{rec}}(\mathbf{x})-O(\mathbf{x})$
penev_A_F00649	022	—	$P(\mathbf{x},\mathbf{x})-P(\mathbf{x}_m,\mathbf{x})\cdot ^{32}$	$P(\mathbf{x},\mathbf{x})-P(\mathbf{x}_m,\mathbf{x})\cdot ^{32}$
penev_A_F00650	022	—	$a_{\mathbf{x}_m}(\mathbf{x})=50\%$	$a_{\mathbf{x}_m}(\mathbf{x})=50\%$
penev_A_F00651	022	—	$a_{\mathbf{x}_m}(\mathbf{x})$	$a_{\mathbf{x}_m}(\mathbf{x})$
penev_A_F00652	022	—	$\left\{ O(x_m)\right\} _{x_m\in \mathcal{M}}$	$\left\{ O(x_m)\right\} _{x_m\in \mathcal{M}}$
penev_A_F00653	022	—	$O(\mathbf{x}_m)$	$O(\mathbf{x}_m)$
penev_A_F00654	022	—	$\mathbf{s}_{\parallel }$	$\mathbf{s}_{\parallel }$
penev_A_F00655	022	—	$\mathbf{s}_{\perp }$	$\mathbf{s}_{\perp }$
penev_A_F00656	022	—	$U_{\mathbf{x}_m}^{\perp }$	$U_{\mathbf{x}_m}^{\perp }$
penev_A_F00657	022	—	$\mathbf{P}_{\perp }=\mathbf{O}^N\mathbf{O}-\mathbf{Q}_{\mathbf{x}_m}$	$\mathbf{P}_{\perp }=\mathbf{O}^N\mathbf{O}-\mathbf{Q}_{\mathbf{x}_m}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
penev_A_F00658	022	—	$\left\langle \text{angle} \mathbf{0} \ \hat{\mathbf{x}}, \ \mathbf{P}_{\perp} \mathbf{0} \ \hat{\mathbf{x}} \right\rangle = P(\mathbf{x}, \mathbf{x}) - P(\mathbf{x}_m, \mathbf{x})$	$\langle \mathbf{O}\hat{\mathbf{x}}, \mathbf{P}_{\perp} \mathbf{O}\hat{\mathbf{x}} \rangle = P(\mathbf{x}, \mathbf{x}) - P(\mathbf{x}_m, \mathbf{x})$
penev_A_F00659	022	—	$\Phi^{*+} \ \phi$	$\Phi^{*+} \phi$
penev_A_F00660	022	—	$\left\{ \mathbf{q}_m \right\}_{x_m \in \mathcal{M}}$	$\{\mathbf{q}_m\}_{x_m \in \mathcal{M}}$
penev_A_F00661	022	—	$U_{\mathcal{M}} \in S$	$U_{\mathcal{M}} \in S$
penev_A_F00662	022	—	$\mathrm{s}_{\setminus } \in U_{\mathcal{M}}$	$\mathbf{s}_{\parallel} \in U_{\mathcal{M}}$
penev_A_F00663	022	—	$s_{\perp} \in U_{\mathcal{M}}^{\perp}$	$s_{\perp} \in U_{\mathcal{M}}^{\perp}$
penev_A_F00664	022	—	$U_{\mathcal{M}}, \left\{ \mathbf{p}_i \right\}_{i=1}^{ \mathcal{M} }$	$U_{\mathcal{M}}, \{\mathbf{p}_i\}_{i=1}^{ \mathcal{M} }$
penev_A_F00665	022	—	$\left\{ \mathrm{q}_m \right\}_{x_m \in \mathcal{M}}$	$\{\mathbf{q}_m\}_{x_m \in \mathcal{M}}$
penev_A_F00666	022	—	$U_{\mathcal{M} \ i}$	$U_{\mathcal{M} i}$
penev_A_F00667	022	—	$\left\{ \mathbf{q}_m \right\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$
penev_A_F00668	022	—	$\mathrm{s}_{\perp} = \mathrm{s} - \mathrm{s}_{\parallel}$	$\mathbf{s}_{\perp} = \mathbf{s} - \mathbf{s}_{\parallel}$
penev_A_F00669	022	—	$\left\{ \mathrm{p}_m \right\}_{\mathrm{m}} \right\}_{\mathrm{x}}_{\mathrm{m}} \in \mathcal{M}$	$\{\mathbf{p}_m\}_{\mathbf{x}_m \in \mathcal{M}}$
penev_A_F00670	022	—	$\left\{ q_m \right\}_{x_m \in \mathcal{M}}, \ s_{\parallel}$	$\{q_m\}_{x_m \in \mathcal{M}}, s_{\parallel}$
penev_A_F00671	022	—	$\left\{ q_m \right\}_{x_m \in \mathcal{M}}$	$\{q_m\}_{x_m \in \mathcal{M}}$
penev_A_F00672		—	$\left\{ \mathbf{q}_m \right\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$
penev_A_F00673		—	$\left\{ \mathbf{q}_m \right\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$
penev_A_F00674		—	$\left\{ O(\mathbf{x}_m) \right\}_{\mathbf{x}_m \in M}$	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in M}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Tables

Identifier	Page	Content (LaTeX source if any)	Scan image
penev_A_TAB_001	014	(no LaTeX source; 391 × 450 px region — see tables.html)	—

Unrecovered image regions — TikZ / table / failed-math candidates

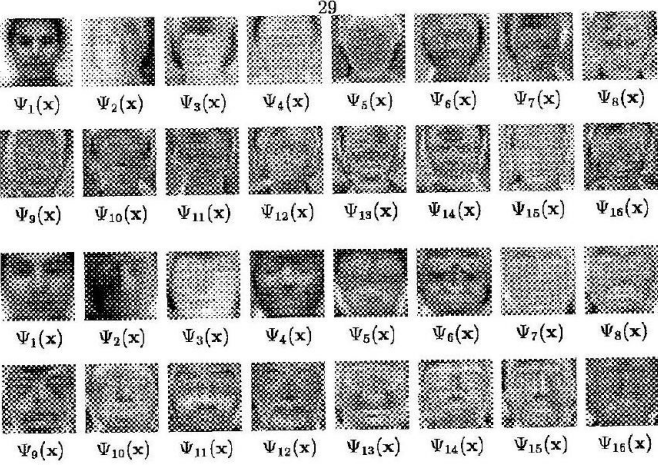
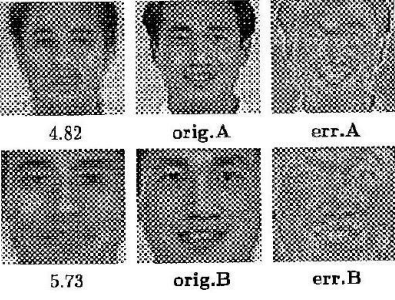
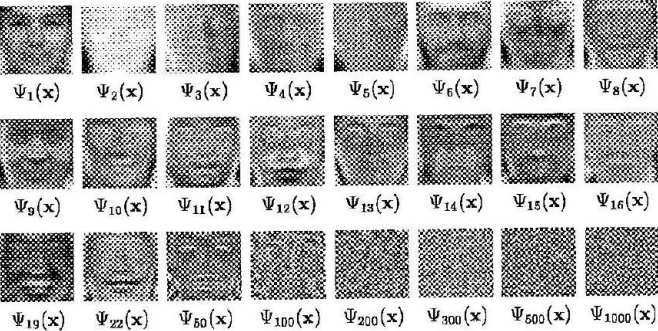
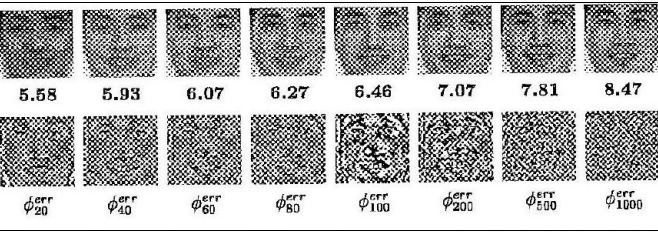
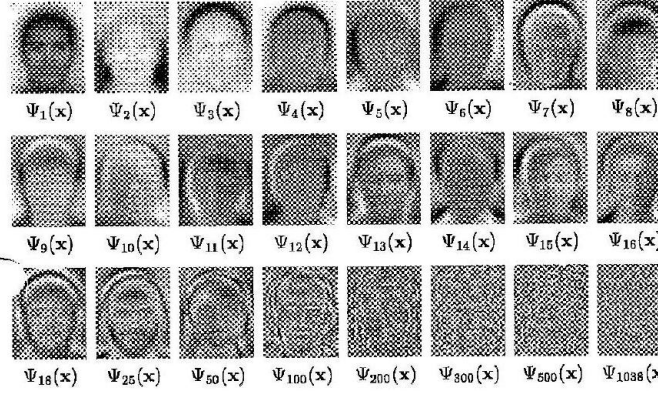
Regions MathPix left as images (no LaTeX). Reconstruct with pdfdrill vision (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

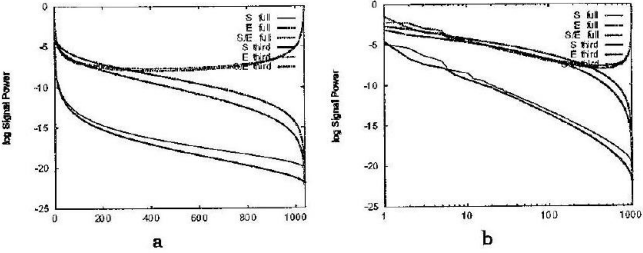
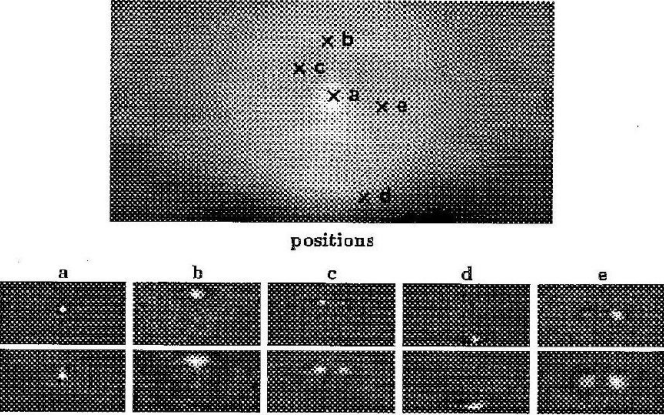
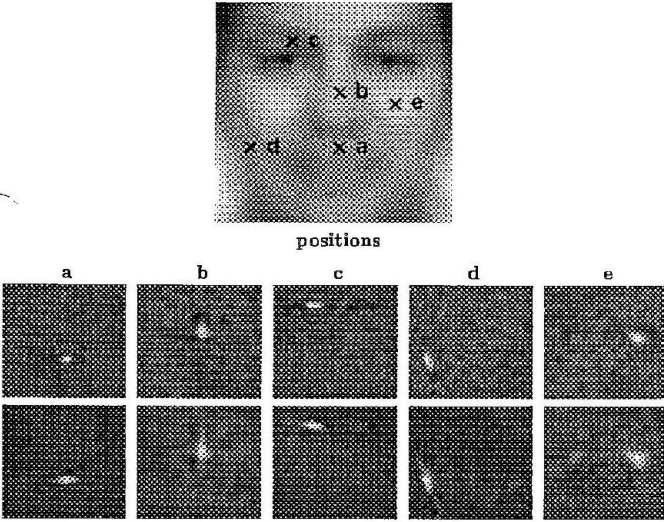
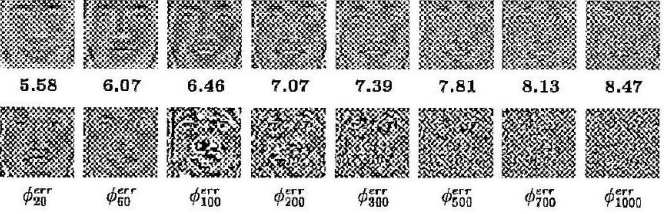
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penev_A_PIC_0003	012		tex.zip (local)

Identifier	Page	Image	Source
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penev_A_PIC_0006	014		tex.zip (local)
penev_A_PIC_0007	014		tex.zip (local)

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